CardioAccess AI - A Multimodal Domain-Adaptive AI System for ECG and Echocardiogram-Based Cardiac Screening

23F-59

Software Requirements Specification (SRS)

BY

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NOVEMBER 2026

Document History

Revision history is maintained below to track changes made to this document by any team member.

Version	Name of Person	Date	Description of Change
1.0	Sahil Kumar, Jawad Ali	October 2026	Initial draft of the SRS document created
2.0	Sahil Kumar	October 2026	Revised draft: updated the distribution list and Sections 1–3 (audience, conventions, background, scope, constraints, operating environment, interfaces, data items exchanged) and Figures 1–3.
2.1 (Revised Draft)	Sahil Kumar	October 2026	Revised the SRS according to the updated CardioAccess AI proposal; updated ECG and echocardiogram analysis, multimodal fusion, datasets, model training, domain adaptation, compression, uncertainty estimation, explainability, performance requirements, and triage functionality.
2.2 (Revised Draft)	Sahil Kumar, Jawad Ali	November 2026	Further revised the SRS by updating the system architecture, functional hierarchy, context and use case diagrams, and use case specifications (UC-01 to UC-07). Refined system requirements, user interactions, security, data privacy, database, storage, deployment, and backend integration to reflect the proposed web-based architecture.
2.3 (Revised Draft)	Sahil Kumar	November 2026	Finalized the use case specifications by completing missing post condition and cross-reference rows in UC-02 to UC-07, refining scenario and alternate scenario descriptions,

Version	Name of Person	Date	Description of Change
			removing development-only activities from the use case diagram, and ensuring consistency across all diagrams, tables, and updated proposal requirements.
2.4 (Revised Draft)	Sahil Kumar	November 2026	Updated the SRS according to the revised proposal by introducing clinician-triggered, on-demand Referral Summary Generation. Updated project scope, architecture, external interfaces, functional hierarchy, use case diagram and specifications, functional and non-functional requirements, security, evaluation metrics, dashboard mock-up, and appendices. Removed outdated statements excluding referral summary generation.

Distribution List

This document will be distributed to the following people after every sign-off:

Name	Role
Prof. Dr. Sher Muhammad Daudpota	Project Supervisor
Dr. Muhammad Ismail Mangrio	Co-Supervisor (Provides technical guidance and review)
Sahil Kumar	Group Lead (Data & ECG Modeling)
Jawad Ali	Team Member - Echo Modeling & Compression Lead
Fatima	Team Member - Integration & Product Lead
Sukkur IBA University Review Committee	Evaluation and Quality Assurance Review
Primary Care Providers (Clinicians / Nurses)	Client / End-User Representative

Document Sign-Off

Once this document is prepared and revised, it will be signed off by the sign-off authority. Any subsequent changes after the first sign-off will require a formal re-sign-off.

Version	Sign-off Authority	Project Role	Sign-off Date
1.0		Supervisor	
2.0		Supervisor	
2.1		Supervisor	
2.2		Supervisor	
2.3		Supervisor	
2.4		Supervisor	

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1. Introduction

1.1 Purpose of Document

This Software Requirements Specification (SRS) document is intended to specify the functional and non-functional requirements for CardioAccess AI. CardioAccess AI is a domain-adaptive, multimodal, and uncertainty aware deep learning based web application for ECG (Electrocardiography) and Echocardiogram based cardiac screening. The system leverages automated ECG analysis, left ventricular assessment from echocardiogram, and multimodal fusion to deliver screening results and triage support particularly in resource-poor primary-care settings. The system is research based screening support system, and not a replacement for the economic clinical diagnosis.

This document serves as a formal reference for the development, design, and validation phases of the project. It outlines the system objectives, scope, constraints, and requirements to ensure that the software developed meets the expectations of the stakeholders.

It will serve as a result of collaboration between the development team, the supervisors, and the evaluation committee, to get all stakeholders aligned on what the system is designed to do.

The main goals of this document are to:

- Define the purpose, scope, and functionality of the CardioAccess AI system.
- Specify software interfaces, evaluation expectations, and dependencies.
- Serve as a foundation for subsequent design (SDS) and implementation activities.

The requirement specification for on-demand Referral Summary Generation feature is described in this document where it says that clinicians who are authenticated to use this feature can generate, review and keep structured summary from the available case information and completed cardiac screening.

1.2 Intended Audience

The target audience for this document is as follows:

Table 1: Intended Audience

Role	Responsibility / Use of this Document
Project Supervisor - Dr. Sher Muhammad Daudpota	Evaluate the project's design alignment, technical feasibility, and adherence to the proposed methodology.
Co-Supervisor - Dr. Ismail Mangrio	Provides technical guidance and review.
Development Team (Sahil Kumar, Jawad Ali, Fatima)	Refer to this document for clear and structured requirements throughout development, testing, and evaluation stages.
Sukkur IBA University Review Committee	Verify that the final system implementation satisfies all stated requirements.
Future Researchers / Students	Use this document as a reference for extending or replicating multimodal cardiac AI research, particularly in domain adaptation, model generalization, uncertainty estimation, and calibration
Clinicians / Nurses	Understand the system's functionalities, patient-case submission process, screening results, triage outputs, and case-history features.
System Administrator	Understand the authentication, access-control, and system-management requirements applicable to the Admin role.

1.3 Document Conventions, Abbreviations, and Acronyms

Document Conventions

- Font Style: Times New Roman
- Font Size: 12 pt (Normal Text), 14 pt (Bold for Sub-Headings), 16 pt (Bold for Headings)
- Heading Numbering: Follows hierarchical IEEE SRS format (e.g., 1, 1.1, 1.2 …)
- Tables and Figures: Center-aligned and sequentially numbered (e.g., Table 1, Figure 1)
- Code/Commands: Monospaced font (e.g., `onnxruntime.InferenceSession()`)

Table 2: Abbreviations

Abbreviation	Full Form / Description
AI	Artificial Intelligence - simulation of human intelligence in machines.
DL	Deep Learning - machine learning using multi-layer neural networks.
ECG	Electrocardiography - recording of the electrical activity of the heart.
Echo	Echocardiogram - ultrasound-based imaging of cardiac structure and function.
LV	Left Ventricle – the heart chamber responsible for pumping oxygenated blood into systemic circulations.
EF	Ejection Fraction – the percentage of blood in the left ventricle ejected with each heartbeat relative to its end-diastolic volume.
PCP	Primary Care Provider - non-specialist clinician in a primary-care setting.
CORAL	Correlation Alignment – an unsupervised domain adaptation method that aligns feature distributions between source and target domains.

Abbreviation	Full Form / Description
ONNX	Open Neural Network Exchange – an open format for representing machine learning models, enabling interoperability and deployment through runtimes such as ONNX Runtime.
SRS	Software Requirements Specification - current document.
SDS	Software Design Specification - detailed design document following this SRS.
RSG	Referral Summary Generation

A complete list of acronyms is provided in Appendix G.

2. Overall System Description

2.1 Project Background

Cardiovascular disease (CVD) is a major global health burden and disproportionately affects low- and middle-income regions where access to specialist cardiac assessment is constrained [1]. This fact is evident in Pakistan: A total of 1,147 cardiologists were certified to treat approximately 240 million people [2]. CardioAccess AI is intended to investigate whether compact, domain-adaptive AI methods can support preliminary cardiac screening in resource-limited settings.[1][2]

Deep learning has demonstrated efficacy in processing electrocardiography (ECG) for cardiac rhythm and ischaemic patterns and echocardiogram (Echo) for cardiac structure and function, left ventricular ejection fraction (EF) [3], [4]. CardioAccess AI merges ECG and echocardiogram assessment as a research-based screening-support prototype. Its outputs are intended to assist preliminary assessment and support referral decisions, not to replace professional clinical judgment or provide a clinical diagnosis [3][4].

However, a structured review of more than 60 studies published between 2019 and 2026 showed that most cardiac AI models are developed and tested on data from a single institution, patient population, and equipment type. Their performance fell dramatically on a distinct case study conducted within a different hospital, country, or population. The CardioAccess AI project attempts to resolve this issue, by reviewing seven gaps highlighted in the literature, five of which relate to the two relevant studies for comparison, Gupta et al. [5] and Nazari et al. [6]:

1. Cross-modal (ECG plus echocardiogram) robustness to domain shift is untested in the comparator studies.
2. Computational complexity and hardware implementability on low-power hardware are not reported in the comparator studies.
3. Confidence calibration under domain shift is not studied in the comparator studies.
4. Label-efficient adaptation is not tested under clinic-realistic data sizes in the comparator studies.
5. Validation is dominated by single-institution datasets.
6. The reviewed comparator studies do not evaluate uncertainty-guided active learning for ECG and echocardiogram adaptation.

7. No human-factors or usability evaluation is included in the comparator studies.

The project is scoped as a research prototype and feasibility study. Its outcome is a working, explainable web application that will undergo a formative usability pilot with volunteer primary-care providers to evaluate usability and perceived trustworthiness.

2.2 Project Scope

CardioAccess AI to generate and augment de-identified ECG and echocardiogram datasets, create efficient ECG and Echo models, evaluate cross-institutional generalization, apply simple domain adaptation and uncertainty-guided active learning, quantify prediction certainty and calibration, and fuse authentic paired ECG and echocardiogram records through learned gated multimodal fusion. These models are incorporated via a browser- accessible web application fronted by React (Vite and Tailwind CSS), with a separate deployment as a FastAPI back end, ONNX Runtime for inference, and the Supabase Cloud for PostgreSQL database management, authentication, and file storage. The two-track research pipeline is shown in Fig. 1.

In Scope:

- Acquisition and preprocessing of seven datasets across the two modality tracks: PTB-XL [7], ARRDB [8], the locally sourced CPEIC/Multan ECG dataset, EchoNet-Dynamic [4], CAMUS [9], MIMIC-IV-ECG [10], and MIMIC-IV-Echo [11]. ECG preprocessing standardizes compatible numerical records to 12 leads and a target sampling frequency of 250 Hz. Records that do not meet these requirements, including single-lead or image-based ECG sources, are handled through separate documented preprocessing and evaluation paths. Patient-level splitting and the four-class taxonomy (Normal / Arrhythmia / Ischemic (MI-related) / Other Abnormal) are applied only where supported by the available dataset labels.
- ECG classification using the proposed SE-ResNet1D architecture, with ResNet1D and Inception1D baseline ablations; PTB-XL [7] is the primary training dataset, while ARRDB [8] and the locally sourced CPEIC ECG-image dataset are candidate external-validation sources, subject to verification of their input formats, lead compatibility, label mapping, digitization feasibility, and data-governance requirements.
- Echocardiogram analysis using a lightweight U-Net for left ventricular segmentation, estimation of the ventricular volume curve, EDV and ESV calculation, EF estimation, and reduced-EF screening. EchoNet-Dynamic [4] is the primary training dataset, while CAMUS [9] is used for external validation. A direct video-to-EF regression model is included as a baseline ablation. The volume-estimation method and its geometric assumptions will be documented.
- Lightweight domain adaptation using limited-data fine-tuning, correlation alignment (CORAL), and a defined test-time adaptation strategy, together with uncertainty-guided active learning compared against random sample selection.
- Uncertainty estimation using Monte Carlo dropout [12] and deep ensembles, prediction-confidence evaluation, calibration assessment, and subgroup (age and sex) performance auditing. Subgroup performance auditing by age and sex will be conducted where the required demographic metadata and sufficient subgroup samples are available.

- A learned gated multimodal fusion network combining ECG-derived and echocardiogram-derived representations, together with relevant echocardiographic measurements. The individual ECG and Echo models are developed using their respective modality-specific datasets, while the fusion network is trained, validated, and tested using genuinely patient-paired MIMIC-IV-ECG and MIMIC-IV-Echo records only. Pairing is based on verified patient-level linkage and predefined temporal correspondence. Patient-level data splitting is applied to prevent data leakage. The fusion model is compared against rule-based fusion and individual modality models using the same held-out paired test patients.
- Model optimization through network Quantization and Knowledge Distillation, ONNX export, and benchmarking of the deployed ECG, Echo and fusion inference components for accuracy, latency, and memory usage on a standard and constrained hardware.
- A browser-accessible research prototype with frontend React (Vite and Tailwind CSS), FastAPI REST backend, ONNX Runtime inference, and Supabase Cloud services for PostgreSQL data persistence, Supabase Authentication, and Supabase Storage. The prototype scenarios are then displaying ECG and echocardiogram screening results and multimodal triage outputs when paired modality data are available, confidence and uncertainty information, and explanation outputs supported by the implemented modes using task completion time, System Usability Scale (SUS), and qualitative trust feedback. The evaluation is a simulation/usability study, not a clinical trial.
- On-demand Referral Summary Generation in the deployed web application. Following a review of the completed screening results, an authenticated clinician may start the creation of a structured summary with the anonymized (pseudonymous) case details and completed ECG, echocardiography, multimodal analysis, etc. outputs, if relevant. An existing pre-built template is used by the FastAPI backend to generate and to store the summary in Supabase PostgreSQL against the case in question, before it is shown to the clinician on the React dashboard.

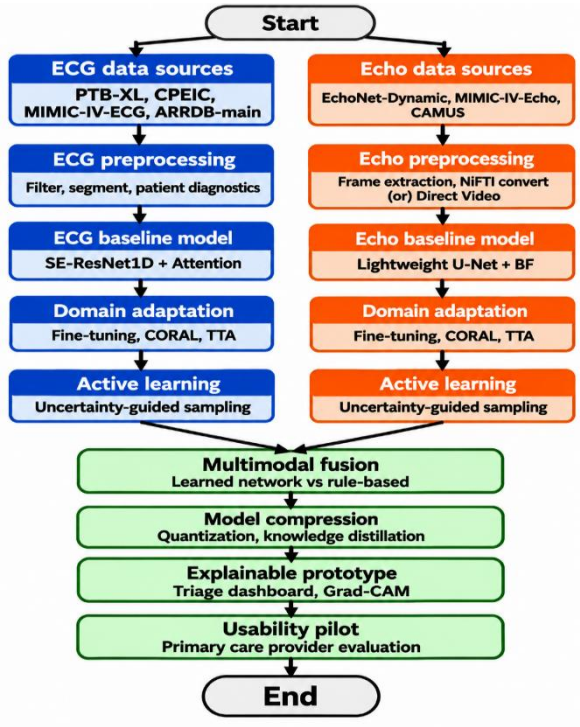


Figure 1: Two-Track Development Pipeline

2.3 Not in Scope

These are explicitly beyond the scope of my CardioAccess AI:

- Clinical validation, clinical trials, or regulatory clearance; use of the system as a medical device for diagnosis or treatment.
- Acquiring new primary level patient data. The project uses publicly available datasets including locally sourced, de-identified research data, subject to applicable data-use permissions and institutional requirements.
- Design of custom or intellectual property hardware, as the project is purely software/AI-based.
- The trained resulted from Gupta et al. [5] and Nazari et al. [6]; are not intended to be faithfully reproduced in entirety, as their functions and numerical accuracy are only necessary to enable the demonstration that the model can take the same type of input and produce a prediction (i.e. a "smoke test" of implementation).
- Federated learning and large-scale foundation-model pretraining; these are considered only as future-work comparisons.
- Synthetic or generative augmentation of target-domain data (future work).
- Automatic referral-summary generation, automatic generation of periodic background processing, automatic escalation of high-priority cases, external referral submission, and external

email or messaging notifications. It is only when an authenticated and authorized clinician explicitly initiates the action through the application that the referral summaries are generated.

2.4 Project Objectives

CardioAccess AI system aims to accomplish the following objectives:

1. Do an analysis of works of the last 7 years (2019–2026) evaluation of AI in cardiology, to identify the extent and consistency of cross-institutional external validation; specifically focusing on ECG based arrhythmia detection and automating ejection fraction estimation from echocardiogram.

2. Train an ECG based cardiac screening classification model (A four-class taxonomy Normal/Arrhythmia/Ischemic (MI-related)/Other Abnormal), and determine its cross-institutional generalization gap through external validation on structurally compatible datasets from other sources.

3. Produce an echocardiogram-based algorithm for accurate segmentation of the Left ventricle/ ejection-fraction estimation, reduced-EF screening and detect its cross-institutional generalization gap through evaluation of an independently sourced dataset.
4. Evaluate lightweight domain adaptation strategies (limited-data fine-tuning, correlation alignment [13], and test-time adaptation) for recovering cross-institutional performance loss in both models, while quantifying the labeled-data requirements.
5. Implement an uncertainty-guided active learning strategy to select informative local samples for adaptation and compare its labeling efficiency against random selection.

6. Quantify model confidence calibration in cross-institutional domain shift and assess if calibration is preserved or degraded after adaptation.

7. Build and evaluate the performance of a learned gated multimodal fusion model that learns to combine ECG-derived and echocardiogram-derived representations and relevant echocardiographic measurements, using patient records where ECG measurements pairs with the adequate echocardiogram reports, and assess its performance and robustness to domain shift against rule-based fusion.

8. Optimize the final ECG and echocardiogram models and the fusion inference component through appropriate compression and deployment techniques, and benchmark accuracy, latency, memory usage, and computational feasibility on hardware representative of primary-care settings with limited resources.

9. Combine the evaluated models and the adaptation, uncertainty, calibration, fusion, and explainability components into a working prototype, and test its usability and perceived

trustworthiness on a small, well-defined pilot with non-specialists (e.g., primary-care workers or providers).

10. On-demand Referral Summary Generation allows authenticated clinicians to generate, read, and save structured summaries of case information that can provide pseudonymous case information and completed cardiac screening results. The completeness, consistency with stored analysis outputs, success of generating and usability of the summary is also measured.

The end measured results on cross-institutional generalization, domain-adaptation recovery, confidence calibration, multimodal fusion, and computational efficiency.

2.5 Stakeholders

Table 3: Stakeholders

Stakeholder	Role / Responsibility
Dr. Sher Muhammad Daudpota	Project Supervisor - oversees system design, ensures alignment with research objectives, and evaluates performance.
Dr. Ismail Mangrio	Co-Supervisor - provides technical guidance and reviews the project's technical direction.
Sahil Kumar	Data & ECG Modeling - Acquisition, pre-processing and patient-level split of ECG datasets; Training and evaluation of ECG baseline models on generalization; Integration of ECG domain adaptation and active learning into systolic dysfunction workflows
Jawad Ali	Echo Modeling & Compression- echo dataset acquisition and format conversion (CAMUS, EchoNet-Dynamic); echo baseline training and generalization testing; model compression and hardware benchmarking (both modalities).
Fatima	Integration & Product – backend development; frontend dashboard development; usability pilot coordination; security/audit layer implementation. Implementation of the on-demand referral summary generation backend function and its React-based summary generation and review interface.
Development Team (shared)	Literature review, weekly decision-log updates, multimodal fusion network, and proposal, mid-year report, final report and defense preparation. Referral summary field mapping, validation of generated summaries, and functional and security testing.
Sukkur IBA University Review Committee	Evaluates project progress, deliverables, and adherence to academic standards.
Primary Care Providers (volunteers)	Participate in the formative usability pilot that evaluates the prototype's usability and perceived trustworthiness.

Stakeholder	Role / Responsibility
End Users (Clinicians / Nurses)	Intended non-specialist users who interact with the web application to submit ECG and/or echocardiogram cases, review screening results, understand explanations and triage outputs, and Access authorized case histories during research usability evaluation.
Admin User	An authorized application role subject to the authentication and access-control permissions defined for the deployed system.

2.6 Operating Environment

The system uses an AI called CardioAccess, which is designed to be run in a browser as the web application interface. The React frontend and FastAPI backend are hosted in distinct environments, and Managed PostgreSQL, authentication and file storage are managed in Supabase Cloud. The research model development pipeline is separate from runtime inference; only final optimized ONNX model artifacts are deployed to the backend inference service.

Hardware Environment:

- This training environment: Google Colab Pro and Kaggle Notebook GPU environments, depending on available computational resources, ECG and echo model development.
- Runtime environment: typical hardware – and a restricted, low-end primary-care system if you wish to test your compressed models against a more limited hardware representation.
- Only nonproprietary, off the shelf hardware is required.

Software Environment:

- Programming Languages: Python 3.x (data pipelines, model development, backend); JavaScript (frontend)
- ML/DL: PyTorch for model development and training; ONNX and ONNX Runtime for model export and inference; supported quantization tools for model optimization; Captum or custom Grad-CAM implementation for ECG explainability.
- Signal and image processing: wfdb, NumPy, SciPy, OpenCV, NiBabel, pydicom, OpenCv
- Backend: Python FastAPI REST APIs; Supabase PostgreSQL, Supabase Authentication, and Supabase Storage
- Frontend: React, Vite, Tailwind CSS, Chart.js, Fetch API
- Operating System: Windows 10/11 or Linux (Ubuntu); Browser: a current version of a Chromium-based browser or Firefox (for the dashboard).
- Testing and collaboration: pytest, Git, GitHub, Google Docs / Notion

- The FastAPI backend supplies the template-based referral summary generation function that uses stored case information and completed analysis to produce referral descriptions; executed through the clinician-triggered program in the React client and allows clinicians to review the summary.

Network Environment:

- The runtime application must be connected to the internet for access to the persistently hosted (but not bundled) React frontend, FastAPI backend and Supabase Cloud services. Disconnected use is not supported.
- Internet access is needed in normal application use to handle user authentication, API communication, database access and file storage/retrieval, as well as for access to a cloud-hosted training resource and credentialed dataset access.
- The React web application uses REST API calls over HTTPS to exchange information with the independently deployed FastAPI backend. The browser also uses Supabase authentication and Supabase Storage for the configured access policies. Requests to the FastAPI backend contain the authenticated user's access token; the backend verifies this before granting access to case files or inference operations.

2.7 System Architecture

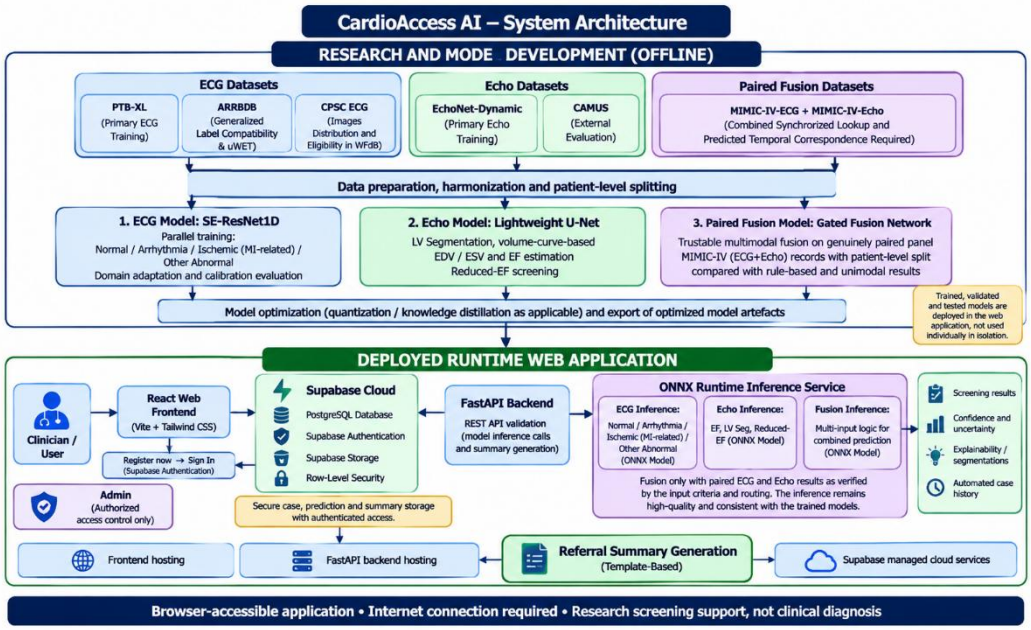


Figure 2: System Architecture

The architecture of the application also allows the on-demand generation of Referral Summaries. An authenticated clinician can request the generation of summary by reviewing the completed screening results on the React dashboard. The FastAPI backend validates request and proper access to requested case, picks up the available case information and analysis outputs from Supabase PostgreSQL, and then supplies them to a pre-built template for use in a structured summary. The resulting summary is stored in Supabase PostgreSQL against the case, and returned to the frontend for review by the clinician. This is an application-level process, and no background scheduler or additional diagnostic model is required.

2.8 System Constraints

- **Software Limitations:** The system relies on specific Python libraries and their compatibility (e.g., PyTorch, ONNX Runtime). Model training depends on the computational resources provided by Google Colab (Pro) and Kaggle Notebooks.
- **Hardware Limitations**: Deployed models must function on low-power, low-end hardware typical of a primary-care setting. This requires model compression techniques such as quantization and knowledge distillation.
- **Data Limitations:** The project utilizes seven de-identified datasets across ECG and echocardiogram tracks. Compatible numerical ECG records are standardized to 12 leads with a target sampling frequency of 250 Hz, while non-compatible formats need separate preprocessing and evaluation documented accordingly. Echocardiogram segmentation annotations are based on available ED and ES frames specific to each dataset. Multimodal fusion involves patient-paired records from MIMIC-IV-ECG [10] and MIMIC-IV-Echo [11] ensuring verified patient linkage and predefined temporal correspondence.
- **Dataset Limitations:** Verification and documentation of identity, size, label format, lead coverage, digitization feasibility, and applicable data-use permissions for the locally sourced CPEIC (Multan) ECG-image dataset must be completed before inclusion in main experiments. If these requirements cannot be met, the dataset will be evaluated separately or excluded from the main ECG experiment.
- **Legal/Regulatory Limitations:** The system is a research prototype, not a clinically validated or regulatory-cleared medical device. No clinical claims are made.
- **User Limitations:** End users are expected to be non-specialist healthcare professionals. Therefore, screening categories, confidence levels, uncertainty indicators, and explanations must be presented in clear language that does not require specialist cardiology knowledge.
- **Schedule and Team Constraints:** The project is structured to be managed by a three-person team across two semesters, with progress tracked through eight monthly milestones and weekly Scrum sprints.
- **Cultural Constraints:** In terms of cultural limitations, the current version of the user interface is available only in English.
- **Environmental Constraints:** Environmentally, the system needs a stable internet connection to operate, as it depends on accessing the browser application, FastAPI backend, and Supabase Cloud services.
- **Off-the-Shelf Component Constraints:** Furthermore, the development leverages off-the-shelf components such as PyTorch, ONNX Runtime, FastAPI, React, Vite, Tailwind CSS, and Supabase Cloud. This use involves adherence to software licensing agreements, ensuring version compatibility, proper service configurations, and compliance with data-use agreements for datasets.
- Referral summary generation is contingent on the availability of corresponding case information, outcomes from a completed analysis. Only the information stored for the selected case is used for the feature, and thus will not change the predictions of the existing model or create new modality results that are missing. Summary generation needs an authenticated and authorized user and an available connection to the deployed FastAPI backend and Supabase Cloud services.

2.9 Abstract System Model

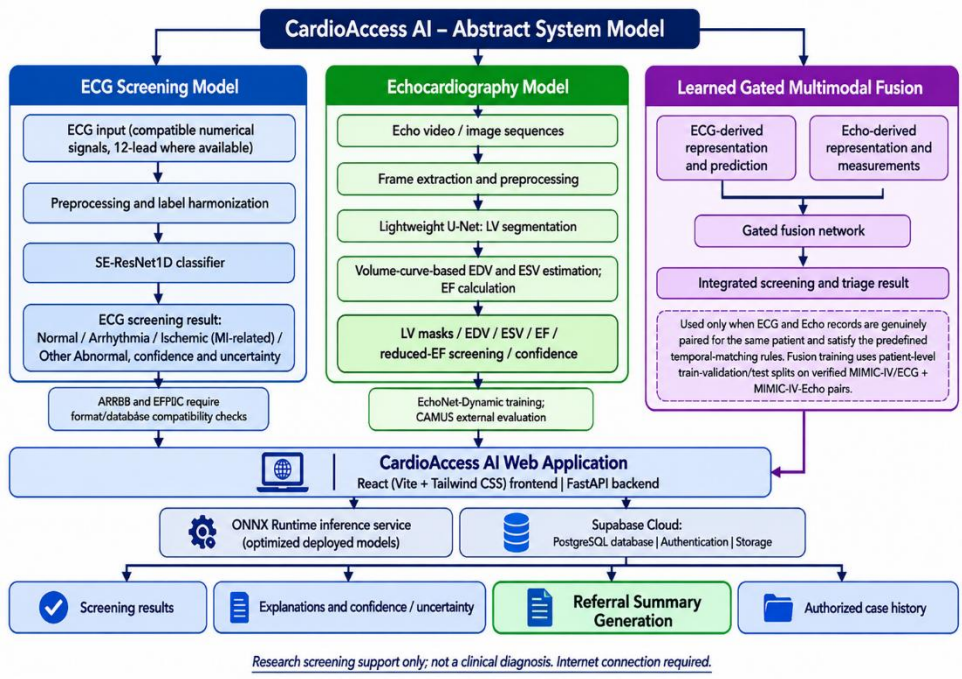


Figure 3: Abstract System Model

2.10 Assumptions & Dependencies

Assumptions:

- All required public datasets and open-source libraries remain accessible.
- Publicly available de-identified datasets may qualify for exemption from institutional ethics approval, subject to the university's applicable research ethics requirements. The locally sourced CPEIC (Multan) dataset is subject to separate verification of data-use permission and any required institutional clearance.
- MIMIC-IV-ECG [10] and MIMIC-IV-Echo [11] provide candidate records for fusion; the number of usable paired records is not assumed until patient linkage, temporal correspondence, and data availability are verified.
- Google Colab (Pro) compute is available for model training.
- Volunteer primary care providers are available for the usability pilot.
- Access to datasets that require credentialed access or data-use agreements, including MIMIC-IV-ECG [10] and MIMIC-IV-Echo [11], is granted to the team and all applicable terms are followed.

Dependencies:

- The system's operation relies on the seven datasets outlined in the project scope, with their accessibility, compatibility, and eligibility determined on a case-by-case basis.
- Dependence on PyTorch and ONNX Runtime for model training, compression, and inference.

- Dependence on the React/Vite/Tailwind frontend, separately deployed FastAPI backend, Supabase PostgreSQL, Supabase Authentication, Supabase Storage, and ONNX Runtime stack for the prototype.
- Dependence on a verified ECG-image digitization approach for the CPEIC dataset, subject to confirmation of its suitability, accuracy, and compatibility with the intended ECG preprocessing workflow.
- Dependence on the published methods of Gupta et al. [5] and Nazari et al. [6] as comparator studies for the smoke-test reproduction.
- Dependence on available pseudonymous case information and successfully stored ECG, echocardiography, or multimodal analysis outputs for generating referral summaries.
- Dependence on the authenticated FastAPI API and Supabase PostgreSQL for retrieving authorized case information and storing generated referral summary records.

3. External Interface Requirements

This section outlines how the CardioAccess AI system interacts with external users, research data sources, computing environments, software components, cloud services, and various communication methods. It describes the necessary interfaces for both the offline model development process and the deployed web application, including functionalities such as data submission, user authentication, model inference, data storage, and dashboard interactions.

The CardioAccess AI system communicates with the following external entities and services:

- **Research Datasets:** Seven specified datasets covering ECG, echocardiogram, and multimodal fusion studies. Access to these datasets is subject to individual dataset restrictions, format requirements, and verification processes.
- **Clinician / Nurse (End User):** These are authorized non-specialist users who submit ECG and/or echocardiogram information, view the relevant screening results, and access confidence levels, uncertainties, explanations, and triage details when available.
- **Development Team and Compute Environment**: A team of developers is responsible for preparing the dataset, training the model, adapting it to specific needs, evaluating its performance, optimizing its efficiency, and deploying it. They use GPU-powered research platforms like Google Colab Pro and Kaggle Notebooks to carry out these tasks.

System Context Diagram

Here's a high-level context diagram that illustrates how the CardioAccess AI system interacts with external entities:

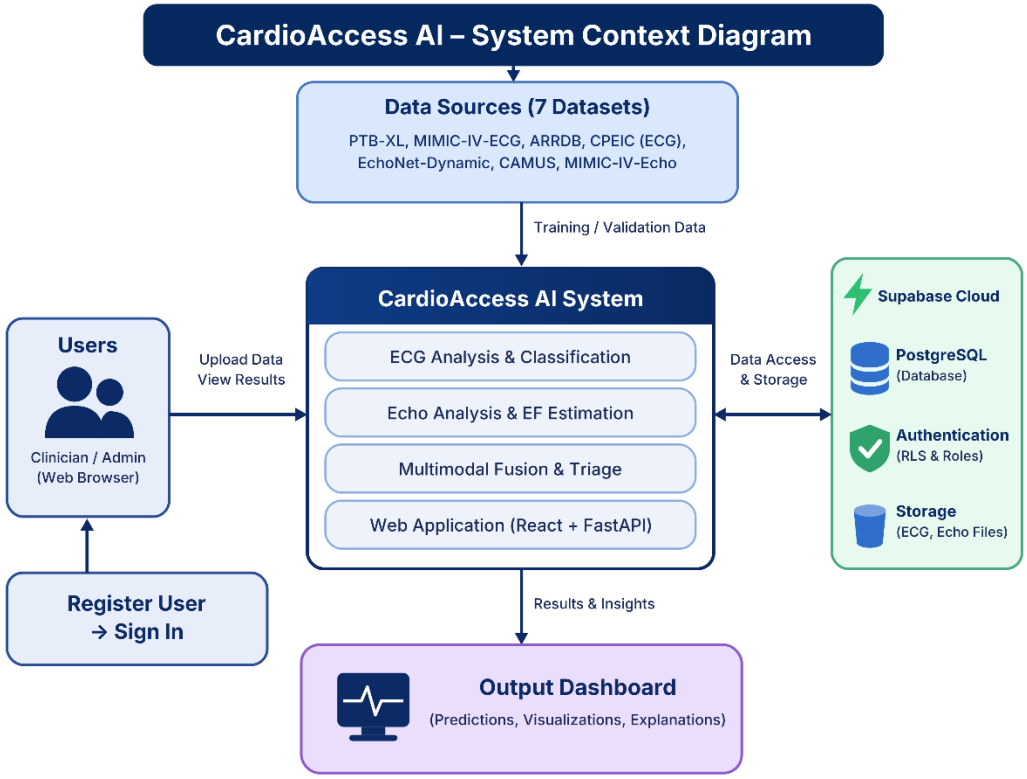


Figure 4: System Context Diagram

3.1 Hardware Interfaces

The project is entirely software- and AI-based, and it does not connect to any proprietary or specialized hardware. The hardware interface refers to the computational environments where the software is trained, tested, and executed. The system does not communicate directly with ECG machines or ultrasound devices; instead, ECG and echocardiogram data are provided in digital format as input recordings.

Table 4: Hardware Interfaces

Hardware Component	Description	Interface Details
Training Compute (Google Colab Pro / Kaggle Notebook)	GPU-accelerated research environments used by the development team for model training, adaptation, and experiments.	Used for research development; not part of the deployed application runtime.
Backend Hosting Environment	Cloud/server environment hosting the FastAPI backend and ONNX Runtime inference service. It is deployed separately from the React frontend and must provide sufficient resources for the optimized model-inference workload.	Separately deployed from the React frontend; exact provider and resource tier are [To Be Confirmed].
Frontend Hosting Environment	Static/web hosting environment serving the React, Vite, and Tailwind CSS application. The frontend is deployed separately from the FastAPI backend and accessed through a supported web browser over HTTPS.	Accessed through a current web browser over HTTPS; exact provider is [To Be Confirmed].
End-User Device	Desktop, laptop, or other browser-capable device used by authorized Clinician/Nurse and Admin users.	Requires internet access and a supported modern browser; no local model installation is required.

3.2 Software Interfaces

This section outlines the software components, tools, and libraries that the CardioAccess AI system relies on for data processing, model development, deployment, and visualization purposes.

Table 5: Software Interfaces

Software Component	Purpose / Functionality	Type of Interaction
Python 3.x	Data pipelines, preprocessing and backend logic.	Executes backend and research scripts.
PyTorch	ECG, Echo and fusion model development, training, adaptation, evaluation and knowledge distillation.	Development-time model training and experimentation.
ONNX Runtime	Optimized runtime inference for exported ECG, Echo, and fusion model artifacts.	Invoked by FastAPI backend for inference.
FastAPI	Backend REST API, authentication-token validation, authorization, file retrieval, preprocessing, inference orchestration, result persistence, and on-demand template-based referral summary generation.	HTTPS REST requests from React; secure server-side integration with Supabase.
React, Vite, Tailwind CSS	Browser-accessible user interface for registration, sign-in, case submission, processing status, screening results, and the clinician-triggered referral summary generation and review interface.	Runs in the user's browser and calls FastAPI/Supabase client services as configured.
Git, GitHub	Version control with a feature-branch workflow across the team.	Stores and versions project code.
Supabase Authentication	Email/password identity management, verification, sign-in, sign-out, and password recovery.	Frontend authentication flow; backend validates issued access tokens.
Supabase PostgreSQL	User centered profiles: roles, pseudonymous cases, file pointers, processing metadata, predictions, probabilities, uncertainty information, model/version metadata, and generated referral summary records connected to the cases themselves.	Access controlled through RLS and backend authorization.
Supabase Storage	Uploaded ECG files, Echo videos, and generated explanation/output artifacts.	Controlled object storage; backend verifies case authorization before retrieval.
wfdb, NumPy, SciPy	ECG parsing, filtering, resampling, and normalization.	Backend preprocessing and research pipeline.

Software Component	Purpose / Functionality	Type of Interaction
OpenCV, NiBabel	Echo video/frame extraction and NIfTI sequence parsing.	Echo preprocessing pipeline.
Captum / custom Grad-CAM	Generation of ECG model explainability outputs where supported by the selected model architecture. Echocardiogram explainability is provided through LV segmentation masks and visual overlays..	Backend/model pipeline produces visual artifacts for dashboard display.
pytest	Automated testing of backend APIs, preprocessing functions, and application workflows	Development and quality assurance.
DICOM reader (if required for MIMIC-IV-Echo)	Reads and processes the selected DICOM echocardiogram files. The specific library will be selected according to the actual file format and preprocessing requirements.	Used only if required by the selected MIMIC-IV-Echo files.
Google Colab (Pro) & Kaggle Notebook (GPU)	GPU-accelerated model training.	Cloud-based execution of training experiments.
ECG-image digitization method (if CPEIC image data is confirmed)	The system transforms suitable ECG trace images into numerical signals once image quality, lead coverage, calibration, and digitization feasibility have been confirmed. This process is exclusively used in the CPEIC preprocessing experiment.	Used in the CPEIC preprocessing experiment only; feasibility and methods.
Monte Carlo Dropout / Ensemble Scripts	During model evaluation, the system generates both confidence and uncertainty estimates. When supported by the deployed inference implementation, it also provides runtime predictions. The exported ONNX workflow must maintain the intended procedure for uncertainty estimation.	Supports uncertainty estimation, calibration evaluation and runtime confidence/low-confidence flags.
PyTorch Quantization Toolkit / ONNX Runtime Quantization Tools	The ECG, Echo, and fusion inference components are optimized where applicable. Knowledge distillation is carried out during model development, while quantization and ONNX export prepare the selected models for deployment.	Compresses the final models before deployment.
Recharts or Chart.js	Displaying confidence scores, calibration curves and case history.	Renders charts in the React dashboard.

Software Component	Purpose / Functionality	Type of Interaction
Axios / Fetch API	Connecting the React frontend to the FastAPI backend for authenticated prediction requests, case operations, referral summary generation requests, and summary retrieval.	Sends and receives HTTPS REST API requests.
Vercel	Hosting the React web application.	Deploys and serves the frontend application.

Versions of libraries and tools will be fixed in the project's requirements file during implementation.

Data Flow:

- Offline model development process: Initially, research datasets are prepared, thoroughly examined, standardized, and separated into training, validation, and testing sets at the patient level when needed. ECG and echocardiogram models are designed and assessed independently using their particular respective data sets. The trained fusion network undergoes training and evaluation using confirmed ECG-Echo paired records of patients. Selected models are then fine-tuned and exported as ONNX-compatible files for deployment purposes.
- Runtime application process: An authorized user logs in via Supabase Authentication and uploads an ECG recording and/or an echocardiogram video through the web platform. Subsequently, the FastAPI backend confirms the access token and authorizes the specific case, pulling permitted files from Supabase Storage. It executes necessary preprocessing steps and runs the appropriate ONNX Runtime model. The fusion model is only triggered when both ECG and Echo inputs are present and matched. Findings, measurements, confidence levels, uncertainty details, and supporting explanations are recorded in Supabase PostgreSQL, whereas associated file assets are stored in Supabase Storage. Lastly, the React dashboard presents the user with approved results.
- Process to create referral summary: When a clinician is satisfied with the completed screening results, they initiate the "Generate Referral Summary" action in the React dashboard. The request is made from the frontend and contains the case reference selected by the clinician in an authenticated request to the FastAPI backend. The backend checks the authentication token and if the user is allowed to access the case. It obtains the case information currently available to it in Supabase PostgreSQL, the completed ECG and echocardiography, as well as fusion outputs when available. It uses a pre-existing template-based function to put all the available information together into a structured referral summary. The generated summary is saved in the Supabase PostgreSQL against the case it belongs to and sent back to the clinician in the React frontend who gets to verify it. The system flags unavailable modality results and it never outputs anything that is not real.

Data Items Exchanged:

The proposed logical data model for Supabase PostgreSQL includes the following tables: user_profiles, which stores information about users, including their authentication details, roles within the application, and profile-related metadata. The cases table contains pseudonymous case identifiers, references to the owning user, timestamps for when the case was created and last modified, modality flags, and the processing status of each case. The uploaded_files table is used to track files associated with a specific case, including the modality of the file, its location in the Supabase Storage system, and related file metadata. ECG results are stored in the ecg_results table, which contains classifications, probabilities,

confidence levels, uncertainty metrics, references to explanations, and the version of the model used. Echo results are captured in the echo_results table, including frame references for end-diastolic and end-systolic phases, segmentation artifact data, EDV, ESV, EF calculations, reduced-EF probability, and uncertainty measures. Fusion results from multiple modalities are kept in the fusion_results table, showing paired case references, combined probabilities, triage categories, modality weights, and the model version used. Lastly, there's a table for processing_events/audit that log event types, their statuses, timestamps, and the actors involved. These are suggested logical components and not yet a finalized physical database structure. Specific details regarding column definitions, foreign keys, data retention policies, and migration strategies are still to be confirmed. Large files and generated media will be stored in Supabase Storage, while PostgreSQL will handle references and structured metadata. The logical data model will also have details of the referral summary records, these will record the created summary content, the relevant case reference, times of creation, and reference to the analysis results tied to the created summary. These records will belong to the case and user with which the records are authorized. Physical schema, including column definitions, foreign keys, and retention policies, will be confirmed during database design. The summary content can be stored as structured text or JSON in Supabase PostgreSQL; an aside from that, separate document storage is needed if PDF download export is integrated.

Table 6: Data Items Exchanged

Direction	Data item	Purpose
Input	Compatible digital ECG recording	Analyzed by the ECG model; 12-lead and 250 Hz standardization is applied where compatible.
Input	Echocardiogram video or supported image sequence	Used for LV segmentation, volume estimation, EF calculation, and reduced-EF screening.
Input	Authenticated summary-generation request and case reference	Identifies the case for which the clinician requests a summary.
Input	Available pseudonymous case information and completed analysis outputs	Supplies the information used by the summary template.
Output	ECG screening class and class probabilities	Provides the four-class ECG result: Normal, Arrhythmia, Ischemic (MI-related), or Other Abnormal.
Output	Structured referral summary	Presents available case details, modality findings, confidence/uncertainty information, and existing triage outcome.
Output	LV segmentation masks and visual overlays	Displays the predicted left ventricular segmentation.
Output	EDV, ESV, EF, and reduced-EF probability	Provides the estimated echocardiographic

		measurements and reduced-EF screening information.
Output	Confidence scores, uncertainty estimates, and low-confidence flags	Communicates prediction confidence and identifies results that may require review.
Output	Triage category (Normal / Needs Review / Urgent Referral)	Provides the integrated screening category when eligible paired ECG–Echo data are available.
Output	ECG Grad-CAM-style explanation, where implemented	Shows relevant ECG regions contributing to the model output.
Shared	ECG and Echo model outputs passed to the fusion network	Allows the fusion network to combine information from verified paired modality inputs.
Shared	Case-log entry and processing metadata	Stores authorized case information, processing status, results, and relevant system events in Supabase PostgreSQL.
Shared	Referral summary record, case reference, generation timestamp, and analysis reference	Stores and links the generated summary to its corresponding case in Supabase PostgreSQL.
Shared	Uploaded files and generated explanation artifacts	Stores the associated files and visual artifacts in Supabase Storage.

3.3 Communications Interfaces

The communication interfaces define the protocols and data-transfer methods used to connect the browser-based React frontend, FastAPI REST backend, Supabase Authentication, Supabase PostgreSQL, Supabase Storage, and ONNX Runtime model-serving components.

Table 7: Communications Interfaces

Communication Type	Protocol / Medium	Purpose	Details / Requirements
Frontend–Backend	REST API over HTTPS (Axios / Fetch API)	Connect the React browser application to the separately deployed FastAPI backend.	Authenticates and provides case references and related input metadata to FastAPI and gets back processing status, ECG/Echo screening results, fusion outputs, where applicable, confidence / uncertainty info, explanation references, and requested referral summaries from an authorized clinician.
Authentication	Supabase Authentication APIs and JWT access tokens over HTTPS	Register/sign in users and authenticate protected application requests.	Email/password, email verification, sign-out, and password recovery; FastAPI validates access tokens before protected operations.
Database Access	Access to Supabase PostgreSQL is managed through either the secure Supabase client or backend operations. Row-Level Security ensures that users can only access data they're authorized for, while the FastAPI backend independently checks permissions for any protected user, case, or data operation. Backend service-role actions should not depend solely on Row-Level Security for protection.	Persist structured profiles, roles, cases, processing metadata, predictions, uncertainty information, and generated referral summary records.	RLS restricts user-scoped access; backend service-role operations must independently authorize every user/case operation.
File Storage	Supabase Storage APIs over HTTPS	Upload and retrieve ECG files, Echo videos, and generated explanation artifacts.	File access is restricted by storage policies and backend authorization; large files remain in object storage, not relational rows.

Communication Type	Protocol / Medium	Purpose	Details / Requirements
Model Serving	Internal backend calls to ONNX Runtime	Run optimized ECG, Echo, and fusion models.	FastAPI handles the preprocessing specific to each modality and performs inference using the ONNX Runtime for both the ECG and Echo models. The fusion model is only triggered when the required ECG and Echo inputs have confirmed patient-level matching and meet predefined time synchronization conditions. Cases involving a single modality are processed using the corresponding model for that modality.
Referral Summary Generation	Authenticated REST API over HTTPS	Request generation and retrieve the generated summary	The React frontend POST a case reference field and the authentication token to FastAPI. The backend checks authorization, looks up the case and analysis details from Supabase PostgreSQL, uses the specific environment to create the summary, logs the summary record, and returns it to the frontend.
Dataset Access	Internet to public/credentialed dataset repositories	Acquire research datasets for model development and evaluation.	Development-time activity; credentialed datasets remain subject to access approvals and data-use agreements.
Cloud Model Development	Internet via Google Colab (Pro) and/or Kaggle Notebook	Train and evaluate research models using GPU resources.	Separate from deployed user-facing inference; raw training datasets are not required on the end-user device.

Communication Security and Synchronization

- The runtime system needs an internet connection to access the frontend, backend, and Supabase Cloud services that are hosted separately.

- Data Transfer: The current version doesn't support real-time streaming. Instead, ECG recordings and echocardiogram videos are submitted as individual files for processing.
- Transport Security: In deployed environments, all communication between the browser and backend, and between the application and Supabase, must use HTTPS/TLS.
- For eligible paired cases, the FastAPI backend manages the inference of ECG and echocardiogram data, and then triggers the fusion network once the required model outputs are available. The final integrated screening result is generated only after the paired-input eligibility checks and required inference steps are completed. Single-modality cases do not invoke the fusion network.
- The current version does not include any external email, FTP, or messaging services. Screening results, low-confidence indicators, and review-related outputs are displayed in the authorized dashboard.
- Referral summary generation is only started by an authenticated user request. In order for this to function FastAPI must first check that the current user is physically able to retrieve or save and return built summary of the case through the request. The summary-generation process is no different to existing HTTPS communication, and exploits Supabase access-control mechanisms instead of scheduled background execution.

4. Functional Requirements

4.1 Functional Hierarchy

The CardioAccess AI project is divided into seven main functional modules, covering research model development, evaluation, and the deployed web application. The research workflow involves data acquisition and preprocessing, development of modality-specific models, domain adaptation, uncertainty estimation, multimodal fusion, and model optimization. The deployed application provides browser-based access, Supabase Authentication, secure file upload, FastAPI request validation and inference orchestration, ONNX Runtime ECG/Echo/fusion inference, Supabase PostgreSQL and Storage persistence, and screening-result presentation. Research model development is performed separately from the runtime application.

- **Module 1 –** Data Acquisition & Preprocessing: Preparation of ECG and echocardiogram research datasets, including ECG signal harmonization, filtering, and 250 Hz resampling where compatible; patient-level data partitioning; dataset-specific mapping to the four ECG screening categories; assessment of CPEIC ECG-image digitization feasibility; Echo frame extraction and preprocessing; CAMUS NIfTI conversion; and preparation of available left ventricular annotations for segmentation.
- **Module 2** – Baseline Modelling & External Validation: This section involves creating the SE-ResNet1D ECG classifier, and comparing it with the baseline models ResNet1D and

Inception1D. Additionally, a lightweight U-Net is developed for segmenting the left ventricle and estimating ejection fraction (EF) based on volume. A direct video-to-EF model is also implemented as a baseline when applicable. The ECG and Echo models are externally validated using designated datasets, allowing for an assessment of their generalization performance.

- **Module 3 –** Domain Adaptation & Active Learning: Here, we investigate techniques such as limited-data fine-tuning, CORAL-based domain adaptation [13], and test-time adaptation depending on the method's feasibility. The module also tests uncertainty-guided sample selection against random selection using similar target data budgets, in order to determine if there's a more efficient approach to resource allocation**.**
- **Module 4 –** Uncertainty, Calibration & Fairness: Predictive uncertainty is estimated using Monte Carlo dropout [12] and deep ensembles wherever possible. We flag predictions that have low confidence levels. To check the calibration of our models, methods like Expected Calibration Error (ECE), Brier score, and reliability diagrams are employed. Furthermore, subgroup analyses based on age and sex are conducted when the dataset includes those attributes and their use is permitted.
- **Module 5 –** Multimodal Fusion: In this part, we develop and evaluate a learned gated fusion network, comparing it to a rule-based fusion baseline. An analysis of modality-reliance weights is carried out as well. The fusion model is trained and tested using verified patient-matched ECG and echocardiogram records from MIMIC-IV-ECG [10] and MIMIC-IV-Echo [11], following the defined criteria for patient-level linkage and temporal matching, ensuring that there are enough suitable paired records for analysis.
- **Module 6 –** Compression & Benchmarking: Optimization of the selected ECG, Echo, and fusion models using the proposed quantization and knowledge-distillation techniques where applicable, export of the deployment models to ONNX-compatible formats, and benchmarking of model size, accuracy retention, inference latency, and memory usage.
- **Module 7 –** Web Application, Referral Summary Generation & Usability Pilot: • Deployed application includes a browser-based React, Vite, and Tailwind CSS frontend. Supabase Authentication provides user registration, email verification, sign-in, sign-out, and password recovery. Supabase Storage is used to store ECG and echocardiogram uploads. FastAPI is used to validate authenticated requests, checks the authorization, and basically orchestrates inference. ONNX Runtime helps us stream the optimized ECG, Echo and fusion models. Supabase PostgreSQL is where we keep the case information, predictions, uncertainty estimates and other metadata. The dashboard gives the screening results, confidence scores, uncertainty scores and explainability results. On-demand Referral Summary Generation function: an authorized clinician can request about case information and completed analysis outputs to get a structured summary. The FastAPI backend creates the summary using the pre-made template, copies it into the case in Supabase PostgreSQL, and returns it back to the dashboard for the clinician to

evaluate. Functional testing and a formative usability pilot will be conducted as part of the application evaluation.

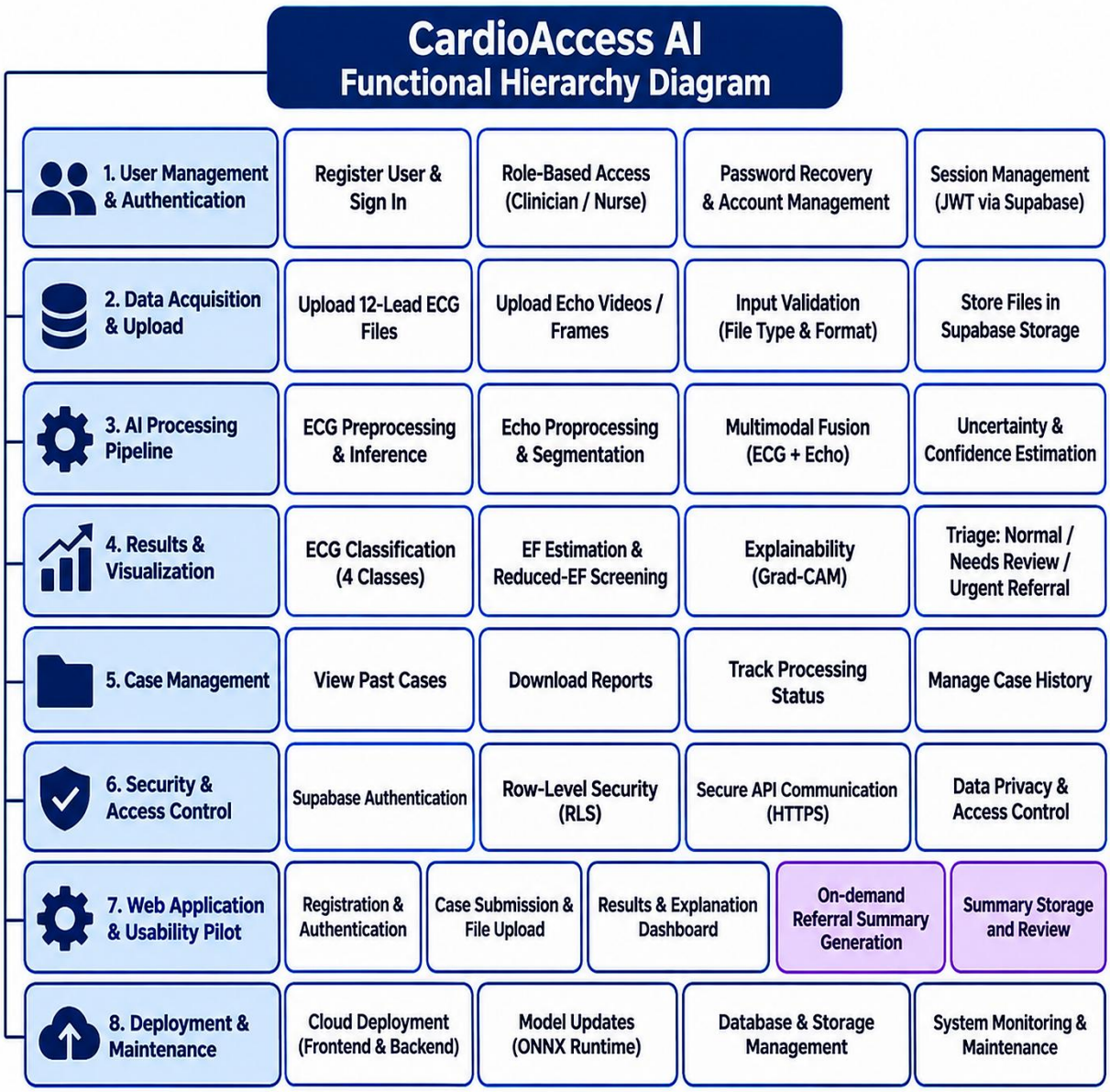


Figure 5: Functional Hierarchy Diagram

4.2 Use Cases

Use Case Diagram

Below is a high-level Use Case Diagram showing the interactions between the main actors and the system:

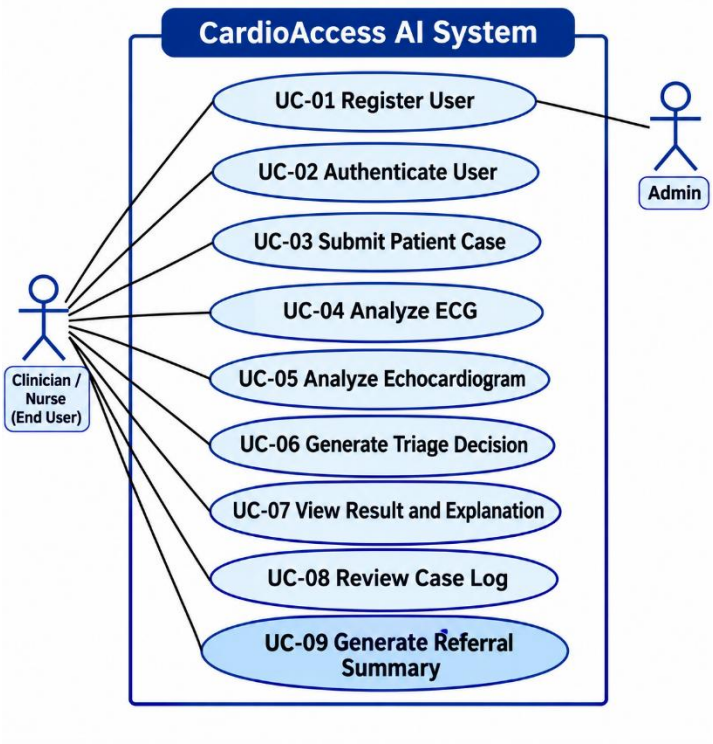


Figure 6: Use Case Diagram

The use case diagram represents the primary interactions between authorized users and the CardioAccess AI system. A new Clinician or Nurse can sign up and then login to use the application as per their role and permissions. Authorized users can submit ECG and/or echocardiograms, trigger the applicable analysis workflows, view screening outcomes and explanations, create triage results, and see authorized case histories. The diagram also depicts one of the use cases, the Referral Summary Generation that will allow an authorized clinician to start an on-demand creation of a structured summary upon review of the screening results completed. This summary was created from relevant case information and stored analysis outputs, a saved one to the relevant case, and put together for the clinician to review. These uses cases here in literature discuss what is actually being implemented in the application and its functions that are available to its users.

4.2.1 Use Case 1: Register User

Use Case Description:

This use case outlines the process by which a new Clinician or Nurse sets up an account on the CardioAccess AI platform via the web application using Supabase Authentication. The user must enter the required registration details, such as an email address and a password. Once this information is submitted, the system forwards the registration request to Supabase Authentication and starts the email

verification process that has been set up. After successfully registering and verifying their email, the user can then log in and access the application.

Table 8: Use Case 01

UC-01: Register User		
Use Case ID:	UC-01	
Actors: New User (Clinician / Nurse)		
Feature: User Registration		
Pre-condition:	• The web application is accessible through a supported browser. • The user does not already have an account with the supplied email address.	
Scenarios		
Step	**Action (Actor)**	**Software Reaction (System)**
1	User opens the CardioAccess AI web application and selects the registration option.	System displays the registration form.
2	User enters the required registration information, including email and password.	System validates the supplied information.
3	User submits the registration form.	System sends the registration request to Supabase Authentication.
4	–	Supabase processes the registration request and creates the account if the information is valid.
5	–	System initiates the configured email verification process and informs the user of the registration status.
Alternate Scenarios:		
2a: If required registration information is missing or invalid, the system displays an appropriate validation message. 3a: If the supplied email address is already registered, the system informs the user that an account already exists. 4a: If registration fails, the system displays an error and does not indicate that the account was successfully created. 5a: If email verification is required but has not been completed, the system informs the user that verification is necessary before accessing protected features.		
Post Conditions		
Step#	**Description**	
1	A new user account is created in Supabase Authentication when registration succeeds.	
2	The email verification process is initiated according to the configured authentication settings.	
3	The user can proceed to sign in after completing the required verification.	

Use Case Cross Referenced:	UC-02 (Authenticate User)

4.2.2 Use Case 2: Authenticate User

Use Case Description:

This use case outlines how a Clinician, Nurse, or Admin can log in to CardioAccess AI via the web application using Supabase Authentication. Once the system checks the provided login details, it creates an authenticated session that grants access based on the user's role. The authentication process includes features like email verification, logging out, and password recovery.

Table 9: Use Case 02

UC-02: Authenticate User		
Use Case ID:	UC-02	
Actors: Clinician / Nurse / Admin		
Feature: User Authentication		
Pre-condition:	• The web application is accessible through a supported web browser. • The user has an account configured in Supabase Authentication.	
Scenarios		
Step#	**Action (Actor)**	**Software Reaction (System)**
1	User opens the CardioAccess AI web application and selects the sign-in option.	System displays the authentication screen.
2	User enters the required email and password and submits the form.	The application sends the authentication request to Supabase Authentication.
3	–	Supabase validates the credentials and returns the authentication result and session token.
4	User accesses the application after successful authentication.	System establishes the authenticated session and displays the authorized application interface according to the user's assigned role.
Alternate Scenarios:		
2a: If the credentials are invalid, the system displays an authentication error and does not grant access. 2b: If account verification is required but incomplete, the system informs the user that verification is required. 2c: If the user requests password recovery, the system initiates the configured recovery process. 2d: If the user signs out, the system ends the active session and returns to the sign-in screen. 2e: If the user selects registration, the system submits the required account details to Supabase Authentication and initiates the configured email-verification process.		
Post Conditions		
Step#	**Description**	
1	The user is authenticated only after successful credential validation.	

2	The authenticated session is established for authorized access.
3	The interface and permitted actions reflect the user's assigned role.
Use Case Cross Referenced:	UC-01 Register User, UC-03 Submit Patient Case, UC-08 Review Case Log

4.2.3 Use Case 3: Submit Patient Case

Use Case Description:

This use case describes how an authenticated Clinician or Nurse submits a compatible digital ECG recording, an echocardiogram video or supported image sequence, or both through the CardioAccess AI web application. The system validates the supplied files and required case information, stores the uploaded files in Supabase Storage, creates the corresponding case record in Supabase PostgreSQL, and initiates analysis for each valid available modality.

Table 10: Use Case 03

UC-03: Submit Patient Case		
Use Case ID:	UC-03	
Actors: Clinician / Nurse (End User)		
Feature: Case Input & File Upload		
Pre-condition:	• The user is authenticated and authorized to submit a case through the web application. • A compatible ECG recording and/or supported echocardiogram video or image sequence is available in digital form.	
Scenarios		
Step#	**Action (Actor)**	**Software Reaction (System)**
1	User opens the dashboard and selects the option to create a new case.	System displays the case input screen.
2	User supplies an ECG signal, an echocardiogram video, or both.	System validates the selected input files and required case information.
3	User submits the case for analysis.	System uploads the files to Supabase Storage and creates the corresponding case record in Supabase PostgreSQL.
Alternate Scenarios:		
2a: If an input is invalid, unsupported, or unreadable, the system displays an error and requests a valid file. 2b: If only one modality is supplied, the system initiates analysis for that modality only. 2c: If no valid input is supplied, the system prevents submission. 3a: If file upload or case-record creation fails, the system reports the failure and does not indicate that the case was successfully submitted.		

Post Conditions	
Step#	**Description**
1	A case record is created in Supabase PostgreSQL when submission succeeds.
2	The submitted files are stored in the authorized Supabase Storage location.
3	Analysis is initiated for the available valid modality or modalities.
Use Case Cross Referenced:	UC-02 Authenticate User, UC-04 Analyze ECG, UC-05 Analyze Echocardiogram

4.2.4 Use Case 4: Analyze ECG

Use Case Description:

This use case describes how the system processes an ECG signal received for a submitted case. The FastAPI backend applies the ECG preprocessing defined by the implemented model pipeline and performs inference using the optimized ECG model through ONNX Runtime. The system generates class probabilities for Normal, Arrhythmia, Ischemic (MI-related), and Other Abnormal conditions, along with confidence levels and uncertainty information. Additionally, it provides a 1D Grad-CAM-style explanation where applicable.

Table 11: Use Case 04

UC-04: Analyze ECG		
Use Case ID:	UC-04	
Actors: System (automated process)		
Feature: ECG Inference		
Pre-condition:	• A valid ECG file has been uploaded and is associated with a case. • The ECG model is available to the FastAPI inference service through ONNX Runtime.	
Scenarios		
Step#	**Action (Actor)**	**Software Reaction (System)**
1	System retrieves the authorized ECG file for the submitted case.	FastAPI validates access to the case and retrieves the file from Supabase Storage.
2	–	The ECG preprocessing pipeline prepares the signal in the format required by the selected model.
3	–	ONNX Runtime executes the ECG model and produces probabilities for the four project categories.
4	–	System calculates the available confidence/uncertainty information and generates the 1D Grad-CAM-style explanation where supported.

Alternate Scenarios:		
2a: If the ECG file is invalid or cannot be processed, the system records the processing failure and reports that analysis could not be completed. 3a: If model inference fails, the system does not present an unverified prediction. 4a: If confidence is below the configured threshold or uncertainty is high, the system flags the output for further review.		
Post Conditions		
Step#	**Description**	
1	ECG class probabilities are available when inference completes successfully.	
2	Confidence/uncertainty information is stored where generated.	
3	The available explanation is associated with the case.	
4	The ECG processing result and status are saved for subsequent triage and display.	
Use Case Cross Referenced:	UC-03 Submit Patient Case, UC-06 Generate Triage Decision	

4.2.5 Use Case 5: Analyze Echocardiogram

Use Case Description:

This use case describes how the system processes an echocardiogram video or supported image sequence associated with a submitted case. The FastAPI backend retrieves the authorized input from Supabase Storage, extracts and preprocesses the required frames, and uses the lightweight U-Net model through ONNX Runtime to segment the left ventricle. Based on the available segmentation outputs and the implemented volume-estimation methodology, the system estimates EDV and ESV, calculates EF, and generates reduced-EF screening and uncertainty outputs where the required measurements are available.

Table 12: Use Case 05

UC-05: Analyze Echocardiogram		
Use Case ID:	UC-05	
Actors: System (automated process)		
Feature: Echocardiogram Inference		
Pre-condition:	• A supported echocardiogram video or image sequence has been uploaded and associated with the case. • The Echo model is available to the FastAPI inference service through ONNX Runtime.	
Scenarios		
Step#	**Action (Actor)**	**Software Reaction (System)**
1	System retrieves the echocardiogram input only after validating the authenticated user's authorization to access the case.	FastAPI validates access and retrieves the video from Supabase Storage.

2	–	System extracts and preprocesses the required video frames.
3	–	The lightweight U-Net model segments the left ventricle in the processed frames.
4	–	System uses the resulting segmentation and defined volume-estimation method to estimate EDV and ESV, where the required measurements are available.
5	–	System calculates EF and generates the reduced-EF screening probability and uncertainty outputs.
Alternate Scenarios:		
2a: If the video is invalid or unreadable, the system reports that the input cannot be processed. 3a: If segmentation fails or the required measurements cannot be obtained, the system marks the affected output as unavailable rather than inventing a measurement. 5a: If uncertainty exceeds the configured threshold, the system flags the result for further review.		
Post Conditions		
Step#	**Description**	
1	LV segmentation outputs are available for successfully processed frames.	
2	EDV, ESV, and EF are stored when the required measurements can be obtained.	
3	Reduced-EF probability and uncertainty outputs are stored when generated.	
4	The Echo analysis status and available segmentation and visualization artifacts are associated with the case.	
Use Case Cross Referenced:	UC-03 Submit Patient Case, UC-06 Generate Triage Decision	

4.2.6 Use Case 6: Generate Triage Decision

Use Case Description:

This use case describes how the system generates a screening triage category from the available modality-analysis outputs. When eligible patient-paired ECG and echocardiogram outputs are available, the system uses the learned/gated multimodal fusion model described in the proposal. For cases without eligible paired inputs, the system uses only a separately defined and supported modality-specific decision pathway. The resulting screening category is Normal, Needs Review, or Urgent Referral. Uncertain outputs are flagged for review, and the available triage result is stored with the case. The output is intended for screening support and does not constitute a clinical diagnosis.

Table 13: Use Case 06

UC-06: Generate Triage Decision		
Use Case ID:	UC-06	
Actors: System (automated process)		
Feature: Multimodal Fusion and Triage		
Pre-condition:	• At least one modality analysis has completed successfully. • The applicable modality-specific or fusion decision logic is available and its input requirements are satisfied.	
Scenarios		
Step#	**Action (Actor)**	**Software Reaction (System)**
1	System collects the available ECG and/or Echo analysis outputs for the case.	System checks the availability, validity, and patient-level pairing eligibility of the modality outputs.
2	–	When eligible paired ECG–Echo outputs are available, the system applies the learned/gated fusion model and generates the fused screening output and modality-reliance information where supported.
3	–	For Echo-based screening, the system applies the defined EF categories: EF greater than 50% = Normal; EF greater than 40% and less than or equal to 50% = Needs Review; EF less than or equal to 40% = Urgent Referral.
4	–	System applies the configured uncertainty-handling rules and flags the result for Needs Review when the relevant uncertainty threshold is exceeded.
5	–	System stores the generated triage output and supporting values in the case record.
Alternate Scenarios:		
2a: If eligible paired ECG–Echo inputs are unavailable; the system does not execute multimodal fusion. It uses a modality-specific pathway only when that pathway has been defined and implemented; otherwise, it displays the available modality result without assigning an unsupported triage category. 3a: If required Echo measurements are unavailable, the EF-based triage category is not generated. 4a: If the output is uncertain, the case is flagged for Needs Review. Note: An ECG-only triage rule must be specified in the implementation before an ECG-only case can receive a triage category; it must not be assumed.		
Post Conditions		
Step#	**Description**	

1	A triage category is stored when the available outputs and implemented decision logic support a decision.
2	Fused cases include the fusion output and modality-reliance information where generated.
3	Uncertain cases are flagged for review.
4	The triage result is available for display in the dashboard.
Use Case Cross Referenced:	UC-04 Analyze ECG, UC-05 Analyze Echocardiogram, UC-07 View Result and Explanation

4.2.7 Use Case 7: View Result and Explanation

Use Case Description:

This use case describes how an authenticated Clinician or Nurse views the screening results for a selected case through the web dashboard. The system retrieves the saved analysis outputs from Supabase PostgreSQL and associated visualization or explanation artifacts from Supabase Storage. The interface shows the available ECG classification results, Echo measurements and segmentation overlays, fusion results when applicable, confidence/uncertainty information, and processing status.

Table 14: Use Case 07

UC-07: View Result and Explanation		
Use Case ID:	UC-07	
Actors: Clinician / Nurse (End User)		
Feature: Dashboard Results and Explainability		
Pre-condition:	• The user is authenticated and authorized to access the selected case. • Analysis results or a processing status are available for the selected case.	
Scenarios		
Step#	**Action (Actor)**	**Software Reaction (System)**
1	User selects a case from the dashboard or case history.	System verifies the user's authorization and retrieves the case information.
2	–	System displays the available ECG class probabilities, Echo measurements and segmentation outputs, EF results, and fused screening information where applicable.
3	User opens the available explanation or visualization.	System displays available Grad-CAM-style explanations, LV segmentation overlays, and modality-reliance information where applicable.
4	–	System indicates incomplete processing, unavailable outputs, or low-confidence/uncertain predictions where applicable.

Alternate Scenarios:	
1a: If the user is not authorized to access the selected case, the system denies access. 2a: If analysis is still processing, the system displays the current processing status instead of completed results. 3a: If an explanation file is unavailable, the system indicates that it is unavailable rather than displaying a nonexistent explanation.	
Post Conditions	
Step#	**Description**
1	The user can view the available screening results for an authorized case.
2	Available explanation outputs and uncertainty indications are displayed with the results.
3	No result is presented as a definitive clinical diagnosis.
Use Case Cross Referenced:	UC-06 Generate Triage Decision, UC-08 Review Case Log

4.2.8 Use Case 8: Review Case Log

Use Case Description:

This use case describes how an authenticated Clinician or Nurse reviews previously submitted cases recorded by CardioAccess AI. The system retrieves case records and their available processing status, predictions, confidence/uncertainty information, and triage outputs from Supabase PostgreSQL. Access to case records and associated files is restricted according to the authenticated user's permissions and the configured database and storage access policies.

Table 15: Use Case 08

UC-08: Review Case Log		
Use Case ID:	UC-08	
Actor: Clinician / Nurse (End User)		
Feature: Case History and Record Review		
Pre-condition:	• The user is authenticated. • The system contains case records accessible to the user.	
Scenarios		
Step#	**Action (Actor)**	**Software Reaction (System)**
1	User opens the case history or case log section of the dashboard.	System retrieves and displays the case records the user is authorized to access.
2	User selects a case from the displayed list.	System retrieves the selected case details and associated analysis results.
3	–	System displays the available case information, processing status, predictions, confidence/uncertainty values, and triage decision.

Alternate Scenarios:	
1a: If no accessible cases are recorded, the system displays an empty case-history view. 1b: If a user attempts to access a case without permission, the system denies the request. 2a: If associated files or results are unavailable, the system indicates their unavailability.	
Post Conditions	
Step#	**Description**
1	The authorized case history is displayed to the user.
2	Details of the selected case are displayed when available.
3	Reviewing a case does not alter its existing stored results.
Use Case Cross Referenced:	UC-02 Authenticate User, UC-07 View Result and Explanation

4.2.9 Use Case 09: Generate Referral Summary

Use Case Description:

The following use case describes the creation of a structured referral summary by an authenticated clinician and uses insights from the review of its completed cardiac screening results. The React frontend provisions the authenticated request with the case reference to the FastAPI backend. The backend verifies the authentication token and the users privileges to access the requested case. The pseudonymous case information, as well as completed ECG, echocardiography, and multimodal analysis outputs, if available, are then fetched from Supabase PostgreSQL. Using a fixed template-based function, the available information is compiled into a structured referral summary without changing the original model predictions or triage outcome. The generated summary is stored in Supabase PostgreSQL against the matching case and shown to the clinician on the frontend.

Table 16: Use Case 09

UC-09: Generate Referral Summary		
Use Case ID:	UC-09	
Actor: Authenticated and authorized Clinician		
Feature: On-demand Referral Summary Generation		
Pre-condition:	• The user is authenticated and authorized to access the selected case. • The selected case has at least one relevant analysis completed successfully. • The corresponding case information and completed analysis outputs are available in Supabase PostgreSQL.	
Scenarios		
Step#	**Action (Actor)**	**Software Reaction (System)**
1	Clinician opens an authorized case and reviews its completed screening results.	System displays the available case information, analysis results, and the "Generate Referral Summary" action.
2	Clinician selects "Generate Referral Summary".	React frontend sends an authenticated request containing the selected case reference to FastAPI.

3	—	FastAPI validates the authentication token and verifies the user's permission to access the selected case.
4	—	Backend retrieves the available case information and completed ECG, Echo, and fusion outputs, where applicable, from Supabase PostgreSQL.
5	—	Backend organizes the available information using the predefined summary template and preserves the original analysis results and triage category.
6	—	Backend saves the generated summary record, corresponding case reference, generation timestamp, and relevant analysis reference in Supabase PostgreSQL.
7	—	FastAPI returns the generated summary to the React frontend.
8	Clinician reviews the generated summary.	System displays the structured summary for clinician review.

Alternate Scenarios:

2a: If the clinician does not have permission to access the selected case, the backend rejects the request and the system displays an authorization error.

3a: If the authentication token is missing, invalid, or expired, the backend rejects the request and does not generate a summary.

4a: If the case has no successfully completed relevant analysis, the system does not generate a summary and informs the user that completed analysis results are required.

4b: If only one modality has completed successfully, the backend generates the summary using that modality's available results and identifies the other modality as unavailable.

4c: If required case information or analysis data cannot be retrieved, the system reports the generation failure and does not save an incomplete or fabricated summary.

5a: If a particular confidence, uncertainty, explainability, or fusion output is unavailable, the summary identifies that information as unavailable rather than assuming or inventing a value.

6a: If summary persistence fails, the system reports the failure and does not indicate that the summary has been successfully saved.

Post Conditions

Step#	Description
1	A structured referral summary is generated from the available case information and completed analysis outputs when the operation succeeds.
2	The generated summary is stored in Supabase PostgreSQL and linked to its corresponding case.
3	The summary is returned to the frontend and displayed for clinician review.

4	The original model predictions, confidence/uncertainty outputs, and triage outcome remain unchanged.
Use Case Cross Referenced:	UC-02 Authenticate User, UC-03 Submit Patient Case, UC-06 Generate Triage Decision, UC-07 View Result and Explanation, UC-08 Review Case Log.

4.3 Functional Requirements Summary

The following table summarizes the functional requirements of the system by module.

Table 17: Functional Requirement Summary

ID	Requirement
4.3.1	**Research and Model Development Requirements**
FR-01	The ECG research pipeline will align compatible 12-lead ECG records and apply a standard sampling frequency of 250 Hz where resampling is needed. Any records that lack full 12-lead coverage will be processed separately or excluded from the 12-lead study and won't be treated as complete 12-lead inputs.
FR-02	The ECG research pipeline will filter and segment suitable ECG signals and divide datasets at the patient level when reliable patient identifiers are available, ensuring no overlap between training, validation, and test subsets.
FR-03	The ECG research pipeline shall map eligible dataset-specific labels to Normal, Arrhythmia, Ischemic (MI-related), and Other Abnormal using a documented label-harmonization procedure.
FR-04	The Echo research pipeline will preprocess echocardiogram videos and frames, convert CAMUS NIfTI sequences into the model input format, and prepare left ventricular segmentation masks from expert annotations. The processing pipeline shall respect the availability of ED and ES annotations in each dataset.
FR-05	The system shall support CPEIC ECG-image digitization only after its image format, lead coverage, labels, source, and clearance are verified; if digitization is infeasible, CPEIC shall be treated as a separately reported experiment or excluded, as documented.
FR-06	The ECG research pipeline shall train and evaluate the proposed SE-ResNet1D classifier and compare its performance with the ResNet1D and Inception1D baseline architectures described in the proposal [3].
FR-07	The Echo research pipeline shall use the lightweight U-Net model for left ventricular segmentation, identify ED and ES frames according to the implemented methodology, estimate EDV and ESV using the defined volume-estimation method, and calculate EF = (EDV − ESV) / EDV × 100 only when valid EDV and ESV measurements are available and EDV is greater than zero

ID	Requirement
FR-08	The Echo research pipeline shall generate reduced-EF screening probabilities and uncertainty estimates using the implemented inference method. For screening-category generation, the defined EF bands shall be EF greater than 50% for Normal, EF greater than 40% and less than or equal to 50% for Needs Review, and EF less than or equal to 40% for Urgent Referral.
FR-09	The Echo research pipeline shall evaluate the direct video-to-EF regression model as a baseline comparison where implemented according to the proposal methodology.
FR-10	The evaluation pipeline shall externally validate ECG and Echo models on the designated datasets and report source-versus-external performance differences and generalization gaps.
FR-11	The research pipeline shall evaluate limited-data fine-tuning and CORAL-based domain adaptation for the ECG and Echo models. Test-time adaptation shall be implemented and evaluated only where its modality-specific procedure, data requirements, and evaluation protocol are defined [13].
FR-12	The research pipeline shall implement uncertainty-guided active sample selection and compare it with random selection at matched target-data budgets.
FR-13	The system shall report Echo adaptation separately for segmentation performance, EF estimation error, and reduced-EF screening performance.
FR-14	The research pipeline shall estimate predictive uncertainty using Monte Carlo dropout and deep ensembles where supported by the model architecture and deployed inference implementation [12]
FR-15	The research evaluation shall assess calibration under in-distribution and domain-shift conditions using ECE, Brier score, and reliability diagrams. For Echo-based reduced-EF screening, calibration shall be evaluated using the corresponding reduced-EF probabilities.
FR-16	The system shall flag predictions that exceed the configured uncertainty threshold and shall not present uncertain outputs as definitive results.
FR-17	The evaluation pipeline shall report subgroup performance by age and sex when the relevant dataset provides those attributes and their use is permitted.
FR-18	The research pipeline shall develop a learned gated multimodal fusion model that combines ECG-derived representations and Echo-derived representations or features using verified patient-paired records.
FR-19	For fused predictions, the system shall provide per-sample modality gate weights or equivalent modality-reliance information when produced by the fusion model.
FR-20	The research evaluation shall compare learned fusion with the proposal's rule-based baseline and the single-modality models.

ID	Requirement
FR-21	The selected ECG, Echo, and fusion models intended for deployment shall be optimized using the proposed quantization and knowledge-distillation techniques where applicable and exported to ONNX-compatible model formats for inference through ONNX Runtime.
FR-22	The project shall benchmark the optimized models for model size, accuracy retention, inference latency, and memory usage on standard and constrained hardware.
4.3.2	**Deployed Application Functional Requirements**
FR-23	The system shall accept ECG recordings and echocardiogram videos or supported image sequences in the file formats defined by the implemented input-processing pipelines. Supported file extensions, codec/container formats, and upload-size limits shall be specified in the system configuration before deployment.
FR-24	he system shall generate the Normal, Needs Review, or Urgent Referral screening category when the applicable modality-specific or multimodal decision policy has been defined and its input requirements are satisfied. Uncertain outputs shall be flagged for Needs Review. The Echo-based EF categories shall follow the defined thresholds: EF greater than 50% = Normal; EF greater than 40% and less than or equal to 50% = Needs Review; EF less than or equal to 40% = Urgent Referral. ECG-only triage shall be generated only after its decision policy has been defined and implemented.
FR-25	The React dashboard will show: the case information, processing status, available ECG class probabilities, echo segmentation overlays and measurements, EF outputs, fused results if available, confidence and uncertainty info, all availability of explainability outputs to me, and the final screening category where relevant.
FR-26	The dashboard will highlight a potential given the confidence level of 0: Low confidence and 2: Needs Review (Urgent Referral) as a low-priority and high-priority screening requirement for a professional review. These flags do not automatically trigger an external referral, notification, or enrichment step
FR-27	The system will store pseudonymous case identifiers, links to authenticated users, timestamps, modality data, processing status, model predictions, class probabilities, confidence and uncertainty details, explanation references, along with model and version metadata in a Supabase PostgreSQL database.
FR-28	The project shall support a formative usability pilot evaluating task completion time, System Usability Scale (SUS) score, and qualitative user trust feedback
FR-29	Authenticated Clinicians and Nurses will be able to access and review only the cases that have been authorized for their specific accounts. This includes all stored predictions, confidence and uncertainty levels, screening classifications, processing statuses, and related case information.
FR-30	New users will be able to create an account by providing an email address and password through Supabase Authentication. Once they've successfully registered and completed their email verification, they can then log in using their registered credentials.

ID	Requirement
FR-31	Supabase Authentication will also support email verification, sign-out functionality, and password recovery options. Before gaining access to any protected features within the application, users must complete the necessary email verification process.
FR-32	The system supports different user roles including Clinician/Nurse end-users and Admins, with access restrictions enforced based on each role. New users will automatically receive the appropriate end-user role through a trusted process, while Admin privileges cannot be assigned via public registration.
FR-33	The FastAPI backend shall validate Supabase authentication tokens and independently enforce user, role, and case authorization for protected API operations.
FR-34	The system shall upload ECG and Echo input files to Supabase Storage and store controlled storage paths/references in the corresponding PostgreSQL case records.
FR-35	The FastAPI backend shall retrieve files from Supabase Storage only after validating the authenticated user's authorization to access the referenced case and file.
FR-36	The system will store structured application and prediction metadata in Supabase PostgreSQL, but it will avoid storing large ECG/Echo binary files directly in relational case rows.
FR-37	Access to the Supabase database will be restricted using Row-Level Security (RLS) policies, ensuring that only authorized users and roles can access the data. Additionally, the backend service-role operations will independently enforce authorization since privileged service-roles might bypass RLS restrictions.
FR-38	Supabase service-role keys, database credentials, and other secrets will be kept exclusively within the backend/deployment secret configuration and will never appear in frontend code.
FR-39	The deployed React frontend and FastAPI backend shall be hosted separately and communicate through secured HTTPS REST endpoints. Supabase Cloud shall provide PostgreSQL, Authentication, and Storage, not frontend/backend hosting or AI inference.
FR-40	The FastAPI backend shall execute the optimized ECG and Echo models and the fusion model, when eligible paired modality inputs are available, through ONNX Runtime. It shall return the available processing status, screening outputs, and associated result information to the frontend.
FR-41	The system shall provide an authenticated and authorized clinician with a "Generate Referral Summary" action for a selected case after at least one relevant cardiac screening analysis has completed successfully.
FR-42	The FastAPI backend will validate a user's Supabase authentication token and verify the user has permission to access the chosen case prior to retreiving case information or a referral summary
FR-43	When processed by the backend, the case information and respective ECG, echocardiography or multimodal analysis results as appropriate will be fetched from Supabase PostgreSQL for the selected case.
FR-44	A structured referral summary will be composed using the available case information and completed analysis outputs, including a predefined template. This summary will not in and of itself diagnose the patient, or change existing model predictions, confidence/uncertainty outputs or triage outcomes.
FR-45	Only finds and/or pseudo-anonymized data from completed modality results will be used by the system in order to build the summary with clear indications of missing modality results or other unavailable information without fabrication of findings.

ID	Requirement
FR-46	The Supabase PostgreSQL will maintain a record of each acceptable referral summary. The record, including its time of creation and relevant analysis references, will be linked to the appropriate case.
FR-47	The Supabase PostgreSQL will maintain a record of each acceptable referral summary. The record, including its time of creation and relevant analysis references, will be linked to the appropriate case.
FR-48	The React Dashboard will display the referral summary to the authorized clinician for review after it's generated and persisted.
FR-49	The system should prompt an appropriate error for authentication, authorization, data retrieval, summary generation, or summary persistence faults and no indication of successful completion of an unsuccessful operation. Logged into the implemented application audit log, relevant summary-generation actions with the case reference, initiating user, generation timestamp, and operation status, are recorded.
FR-50	The system should not auto-generate referral summaries on completion of analysis, by scheduled background execution, or by a scheduler. Summary generation is allowed only upon request from an authenticated and enabled clinician. The system shall not automatically raise urgent cases or submit external referrals.

5. Non-Functional Requirements

This section defines the qualitative characteristics and constraints that affect the performance, safety, security, and documentation of the CardioAccess AI system.

5.1 Performance Requirements

The CardioAccess AI system is designed to investigate reliable cardiac screening performance on low-cost hardware. Because CardioAccess AI is a research prototype, its performance requirements are expressed as evaluation targets and measurable outputs:

Table 18: Performance Requirements

Parameter	Requirement	Description
ECG classification	Report AUC, sensitivity, specificity, and macro-F1 on PTB-XL and external datasets.	The proposal lists a target AUC ≥0.90 in-distribution; results remain pending evaluation.
Echo LV segmentation	Report Dice and IoU on annotated ED and ES frames.[4]	The proposal lists a target Dice ≥0.85 in-distribution; results remain pending evaluation.
Echo EF estimation and reduced-EF screening	Report EF MAE/RMSE and screening AUC, sensitivity, specificity, and PR-AUC using confirmed thresholds.	Volume-estimation assumptions and threshold boundaries require confirmation; do not interpret segmentation alone as clinical volume validation.

Parameter	Requirement	Description
Cross-institutional generalization	Report absolute and percentage changes between source and external datasets.	Applies independently to ECG and Echo models. Performance shall be compared before and after adaptation using the same held-out evaluation data.
Domain adaptation	Report adaptation recovery under proposal-defined target-data budgets and CORAL settings.[13]	No unsupported runtime latency target is imposed.
Confidence calibration	Report ECE, Brier score, and reliability diagrams in-distribution and under domain shift.	Echo calibration uses reduced-EF probability.
Active learning	Compare uncertainty-guided and random sample selection at matched labeling budgets.	Report comparative recovery/label efficiency.
Multimodal Fusion	The learned fusion model shall be evaluated against rule-based fusion and each single-modality model.	Fusion is trained/evaluated only on genuinely patient-paired ECG–Echo records with verified linkage and temporal correspondence.
Inference Efficiency	Accuracy retained, inference latency, and memory shall be benchmarked on standard and constrained hardware.	Specific numerical latency/memory targets are to be confirmed after selecting the benchmark device.
Connectivity	Normal deployed operation shall require internet connectivity to reach the frontend, FastAPI backend, and Supabase Cloud services.	The deployed application depends on online access to its separately hosted components.
Modularity	The deployed React frontend, FastAPI backend, Supabase services, and ONNX Runtime inference layer shall remain separately identifiable components.	Research model training and adaptation are separate from runtime inference.

Parameter	Requirement	Description
Usability	Usability shall be measured in the formative pilot.	Task completion time, System Usability Scale (SUS) score, and qualitative trust feedback.
Referral Summary Generation	Report summary completeness, consistency with stored analysis outputs, successful generation rate, and task-completion time.	Evaluates whether the system generates complete and accurate summaries from available case information, preserves existing screening results, handles unavailable modality outputs correctly, and supports an efficient clinician review workflow.

5.2 Safety Requirements

The CardioAccess AI system provides preliminary research-oriented screening outputs. It is not an autonomous diagnostic system, a clinically validated medical device, or a replacement for professional assessment. Safety requirements highlight the importance of communicating uncertainty, handling incomplete inputs, understanding the limits of explainability, and maintaining human oversight.

Table 19: Safety Requirements

Safety Area	Requirement / Safeguard
Intended Use	The system is a research prototype and feasibility study. It is not a clinically validated or regulatory-cleared medical device, and no clinical claims shall be made.
Uncertainty Handling	When the model's predictions exceed its validated uncertainty threshold, they should be marked for human evaluation instead of being delivered with unwarranted confidence. For echocardiogram cases where certainty is low, a "Needs Review" status must be assigned regardless of the ejection fraction category.
Review Flagging	Screening results designated as Low Confidence, Needs Review, or Urgent Referral must be clearly identified for healthcare professionals to examine. These are merely labels used in the screening process and do not mean the system will independently reach out to external services or make any referral decisions on its own.
Human Oversight	The system is intended to assist authorized clinicians and nurses in their screening tasks. However, the final clinical judgment and decision-making responsibilities remain with properly qualified healthcare professionals.
Certifications and Regulations	The project shall not claim medical-device certification, regulatory clearance, or clinical validation. Its research-prototype status shall be clearly communicated to users.
Explainability	ECG predictions should be presented alongside Grad-CAM-style visual explanations when available. Echocardiogram results should also include visual components such as LV segmentation overlays or similar explanations to aid users in reviewing the model's output. It's important to note that these visuals are meant for clarification, not as evidence of the accuracy of the clinical assessment.
Fairness	Performance shall be reported by age and sex before and after adaptation to detect demographic disparities.
Auditability	Every prediction, confidence score and system action shall be recorded in the case log.
Usability Pilot	The pilot is a simulation study with volunteer participants and is explicitly not a clinical trial.
Referral Summary Accuracy	Generated summaries hide information not relevant to the case or in the analysis outputs. Missing modality findings or measurements are marked as unavailable and should not be reported numerically.

Safety Area	Requirement / Safeguard
Clinician Review	Referral summary shares relevant screening information to facilitate safe and efficient referral support and is not designed to replace the clinical judgment or form the basis of an independent diagnosis, treatment recommendation or referral decision.

Note: Please note that CardioAccess AI is a research-based preliminary screening tool. The AI-generated results should not be considered as definitive diagnoses, nor should they replace professional medical evaluation. Additionally, the tool has not undergone clinical validation or received regulatory approval.

5.3 Security Requirements

Security controls will be employed across Supabase authentication, FastAPI, PostgreSQL RLS, Supabase Storage, and deployment settings as CardioAccess AI processes uploaded ECG and echocardiogram files, pseudonymous case records, AI generated screen results, and clinician requested referral summaries. All protected operations involving case information & generated summaries require backend and database layer authentication and authorization.

Table 20: Security Requirements

Security Aspect	Requirement Description
Authentication	Supabase Authentication shall provide email/password registration and sign-in, email verification, sign-out, and password recovery.
Token Validation	FastAPI shall validate Supabase JWT/access tokens on every protected API request and reject invalid, expired, or missing credentials.
Authorization and Roles	The system will ensure that Clinicians, Nurses, and Admins have appropriate access at both the backend and database levels, based on their assigned permissions. It is important to note that frontend visibility settings should not be used as a substitute for proper authorization.
Database Access Control	Supabase PostgreSQL Row-Level Security policies shall restrict access to authorized user/case records. Backend service-role operations shall independently enforce authorization.
File Access	Supabase Storage objects shall be accessible only through authorized policies or backend-mediated access after case ownership/role checks.
Secrets Management	Supabase service-role keys and all privileged credentials shall remain backend-only and shall not be embedded in React bundles or exposed to browsers.
Transport Security	All deployed frontend/backend and backend/Supabase communication shall use HTTPS/TLS.
Data Minimization	The system shall use pseudonymous case identifiers and store only the structured metadata and files required for the proposal-defined research workflow.
Audit Trail	The system shall record relevant case-processing status, predictions, timestamps, model/version metadata, and implemented user or administrative actions.
Data Governance	Only de-identified research datasets can be used, provided they comply with relevant source permissions and data-use agreements. Before using any dataset, it is essential to confirm its access conditions and ensure its suitability for use with CPEIC.
Session Handling	The application shall use Supabase Authentication session mechanisms and shall not expose access tokens or privileged credentials in logs or publicly accessible storage.
Referral Summary Authorization	The FastAPI backend is required to confirm that the authenticated user's permissions for the selected case before it retrieves its information, generates a summary, or returns a stored summary.
Referral Summary Data Protection	Summary data must be linked to its authorized cases and protected using the appropriate RLS policies on the PostgreSQL backend and backend authorization checks. It must never be readable by an unauthorized caller

5.4 User Documentation

Comprehensive documentation will be provided alongside the CardioAccess AI project to support evaluation, reuse, and future development. This documentation package will include:

Table 21: User Documentation

Document Type	Description
Project Proposal	Revised FYP proposal describing the problem, research gaps, methodology, and plan.
Software Requirements Specification	This document, defining the functional and non-functional requirements of the system.
Mid-Year Report	Progress report covering the work completed up to the mid-year checkpoint.
Final Report	Final project report and thesis document presenting the methodology, results and evaluation.
Decision Log and Risk Register	Shared weekly log of decisions and risks maintained by the team during the sprints.
Defense Material	Material prepared for the final project defense.
User Guide / Application Manual	Describes user registration and sign-in, case submission, screening-result review, on-demand referral summary generation, and summary review.

6. References

This section lists the documents, articles, and sources that were referenced in preparing this Software Requirements Specification (SRS) document.

Table 22: References

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Note: All references above can be accessed from reputable scientific repositories such as PhysioNet, IEEE Xplore, Nature, and arXiv.

7. Appendices

This section contains supplementary materials such as figures, charts, and additional details that provide extra information to support the main body of the SRS document. While not necessary for an immediate understanding of the system requirements, these elements serve as useful reference material for developers, evaluators, and future researchers working on extending the project.

Appendix A – Compute and Deployment Environment Details

Table 23: Appendix A - Compute and Deployment Environment Details

Component	Specification / Description
Training Environment	Google Colab (Pro) and/or Kaggle Notebook with GPU acceleration for research training and experiments.
Model Format	Compressed ECG and Echo models exported to ONNX for deployed inference.
Inference Runtime	ONNX Runtime integrated with the separately deployed FastAPI backend.
Frontend Deployment	React application built with Vite and styled using Tailwind CSS; hosted separately from backend.
Backend Deployment	Python FastAPI REST service deployed separately; containerization/deployment provider and resource tier are [To Be Confirmed].
Cloud Data Services	Supabase Cloud provides PostgreSQL, Supabase Authentication, and Supabase Storage; it does not host the complete application or run model inference.

Notes:

- No proprietary or specialized hardware is required.
- The deployed system needs only the compressed model artifacts, not the raw training datasets.
- In terms of systems architecture, the FastAPI backend also does the start of the end-user: in the case of a clinician, it has the Summit-based referral summary generation which the Supabase PostgreSQL sees its summary records and connects per cases, without a separate scheduling service or any external AI inference hardware to do the job.

Appendix B – Dataset Specifications

Table 24: Appendix B - Dataset Specifications

Dataset	Modality	Role in Project	Description / Notes
PTB-XL [7]	ECG	Primary ECG training dataset	Public research data set. Its diagnostic taxonomy will be the basis for a new approach to four classes of ECG screening, as described in the proposal. The exact 12-lead pre-processing and split configuration will be documented
ARRDB [8]	ECG	Candidate external ECG validation dataset	It is the VitalDB Arrhythmia Database, which the proposal also experiments with. The proposal offers no lead coverage input compatibility with the 12-lead model; check before use, and do not default to 12-lead coverage
CPEIC / Multan	ECG image / locally sourced ECG	Candidate external validation or separately reported experiment	CPEIC locally sourced ECG dataset appears in proposal, however there are other local ECG datasets listed in an inconsistent manner. The source institution, identity, number of datasets, lead coverage, image/signal format, labels, access and ethical clearance should be verified.
EchoNet-Dynamic [4]	Echo video	Primary Echo training dataset	Public research dataset. Expert LV tracings are specified for ED and ES frames; the model is applied according to the proposal's frame/volume methodology.
CAMUS [9]	Echo sequences (NIfTI)	External Echo validation dataset	Public research dataset; sequences are converted into the common frame representation. Confirm exact inference input and annotation mapping.
MIMIC-IV-ECG [10]	ECG	Candidate multimodal fusion dataset	Credentialed/access-controlled research dataset. Candidate records must be linked to MIMIC-IV-Echo using verified patient-level identifiers and suitable temporal correspondence; do not assume every ECG has a paired Echo.

Dataset	Modality	Role in Project	Description / Notes
MIMIC-IV-Echo [11]	Echo	Candidate multimodal fusion dataset	Credentialed/access-controlled research dataset. Candidate records must be linked to MIMIC-IV-ECG using verified patient-level identifiers and suitable temporal correspondence; data-use requirements apply.

All datasets are used as de-identified research data under their respective access terms (public, credentialed or restricted) or as locally sourced data; no primary patient data is collected.

Appendix C – Software Tools and Libraries

Table 25: Appendix C - Software Tools and Libraries

Tool / Library	Purpose
Python 3.x	Model development, research data pipelines, and FastAPI backend logic.
PyTorch	Training ECG, Echo, and fusion models; domain adaptation, uncertainty, and compression experiments.
ONNX Runtime	Optimized model inference in the FastAPI backend.
FastAPI	REST API, request validation, authorization, preprocessing, inference orchestration, result persistence, and on-demand template-based referral summary generation.
React, Vite, Tailwind CSS	Browser-based frontend, build tooling, styling, screening dashboard, and clinician-facing referral summary generation and review interface.
Supabase Authentication	Email/password authentication, email verification, sign-out, and password recovery.
Supabase PostgreSQL	Structured application records, prediction metadata, pseudonymous case information, and generated referral summary records linked to corresponding cases, with Row-Level Security.
Supabase Storage	ECG/Echo uploads and generated explanation/output artifacts.
Captum / custom Grad-CAM	Generating proposal-supported explainability outputs.
wfdb, NumPy, SciPy	ECG parsing, filtering, resampling, and normalization.
OpenCV, NiBabel	Echo video/frame extraction and NIfTI parsing.
pytest	Backend and data-pipeline unit/integration testing.
Git, GitHub	Version control and collaboration.

Tool / Library	Purpose
Google Colab (Pro), Kaggle Notebook	GPU-accelerated model training and experiments.
Git, GitHub	Version control and collaboration
Google Docs / Notion, Microsoft Word	Decision log, risk register, and formal documentation
PyTorch quantization toolkit, ONNX Runtime quantization tools	Post-training quantization of the final models (knowledge distillation is implemented in PyTorch)
DICOM reader (e.g., Pydicom)	Reading MIMIC-IV-Echo DICOM files
ECG-image digitization method (existing published method, to be selected)	Converting CPEIC ECG images into signals
Operating System (Windows 10/11 or Linux (Ubuntu))	Host environment for development, backend and dashboard

Appendix D – Model Evaluation Metrics

The following metrics are used to assess the performance of the trained models and the prototype:

Table 26: Appendix D - Model Evaluation Metrics

#	Component	Metric(s)	What It Demonstrates
1	ECG baseline model	AUC, sensitivity, specificity (per class)	In-distribution diagnostic performance on PTB-XL [7]
2	Echo segmentation	Dice coefficient, IoU (end-diastole and end-systole frames)	Accuracy of LV segmentation on EchoNet-Dynamic [4]
3	Echo EF estimation	EF MAE, RMSE, R^2, Bland–Altman agreement	Accuracy of the EF value against the labeled EF
4	Echo reduced-EF screening	AUC, sensitivity, specificity, PR-AUC (EF < 50% and ≤ 40%)	Ability to flag reduced function; PR-AUC is used because classes are imbalanced
5	Cross-institutional generalization	Absolute and percentage drop in the above metrics (source vs. external dataset)	Quantifies the real-world generalization gap for both modalities
6	Domain adaptation (fine-tuning)	Recovery curve at 5%, 10% and 25% target-data budgets	How much labeled local data is needed to close the gap

#	Component	Metric(s)	What It Demonstrates
7	Recovery achieved with zero target labels; evaluates the feasibility of CORAL-based domain adaptation.	Recovery achieved with zero target labels	Feasibility of adaptation when no local labels are available
8	Active learning vs. random sampling	Recovery curve comparison at matched data budgets	Whether uncertainty-guided selection improves labeling efficiency
9	Confidence calibration	ECE, Brier score, reliability diagrams, in-distribution vs. out-of-distribution	Whether model confidence remains trustworthy after domain shift
10	Multimodal fusion	Accuracy/AUC of learned fusion vs. rule-based fusion vs. each single-modality model	Whether joint ECG–echo reasoning improves robustness
11	Model compression	Accuracy retained (%), inference latency, memory footprint (standard vs. constrained hardware)	Deployability on low-power, primary-care-representative hardware
12	Subgroup fairness	Performance stratified by age and sex, before and after adaptation	Whether adaptation introduces demographic performance disparities
13	Usability pilot	Task completion time, System Usability Scale (SUS) score, qualitative trust feedback	Practical usability and perceived trustworthiness among primary care provider volunteers
14	Referral Summary Generation	Information completeness, consistency with stored analysis outputs, correct triage representation, successful generation rate, and task-completion time	System completeness, consistency with stored analysis outputs, fully linked triage result, success rate, and time to fulfilling the task Assesses the system to producing complete, accurate, correctly triaged summaries, provides modality that is not available, keeps predictions and triage, and assists clinicians to have reviews.

Appendix E – User Interface Mock-up

Dashboard Features (React):

- Triage result (Normal / Needs Review / Urgent Referral)
- Confidence scores and low-confidence flagging
- Grad-CAM-style explainability overlays for ECG and echo predictions
- Modality reliance (gate weights) for fused predictions
- Calibration curves and case history charts
- Authorized case history, review flags, and processing status
- A clinician-facing "Generate Referral Summary" action displayed after at least one relevant analysis has completed successfully.
- A structured referral summary review interface displaying available case information, completed modality findings, confidence and uncertainty information, and the existing triage outcome.
- A summary generation status and error message area indicating whether the summary was successfully generated and stored.

Mock Layout Example:

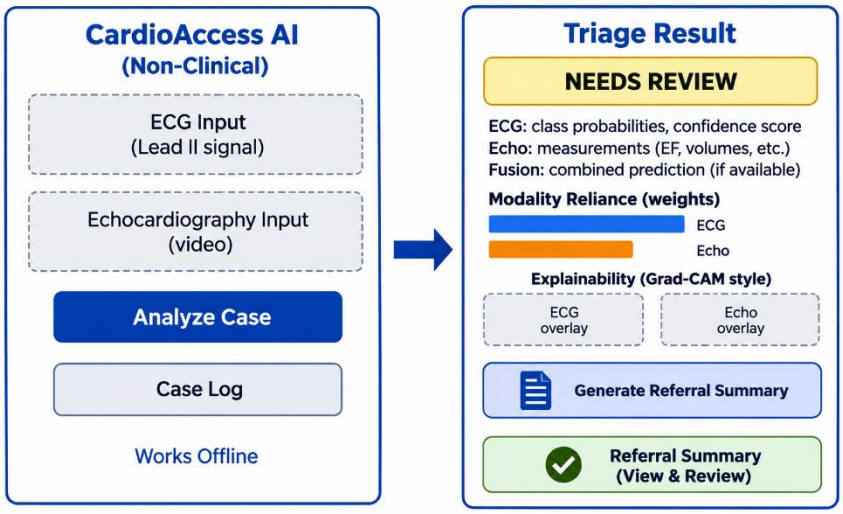


Figure 7: Dashboard Interfaces (Illustrative Mock-up)

The mock layout shows the case input screen, the dashboard and screening result screens. It similarly includes the clinician postcode-triggered referral summary generation and review interface in UC-09. Case history, triage information and screening explanation are found in UC-06, UC-07, and UC-08, respectively.

Appendix F – Ethical and Research Compliance

- Model development uses only publicly available, credentialed or restricted-access de-identified research datasets, and locally sourced data; no primary patient data is collected, so no institutional ethics board approval is required for the core research on these datasets. The CPEIC (Multan) dataset is subject to clearance verification.
- Use of credentialed or restricted datasets, including MIMIC-IV-Echo [11], shall comply with the applicable data-use agreements.
- The source institution, size, label format and ethical-clearance status of the CPEIC (Multan) dataset will be documented before it is used.
- The usability evaluation is a simulation study with informed, voluntary participants and is explicitly not a clinical trial; no clinical claims are made to participants.
- The system is a research prototype and is not a clinically validated or regulatory-cleared medical device.
- The reproduction of Gupta et al.'s [5] and Nazari et al.'s [6] techniques is a smoke test of implementation feasibility, not a claim of reproducing their full reported results
- Referral summaries will be created and examined using de-identified or fake case information and model outputs. These are intended as structured referral-support documentation for a clinician to review and are not independent clinical diagnoses or treatment recommendations.

Appendix G – Acronyms and Definitions

Table 27: Appendix G - Acronyms and Definitions

Acronym	Definition
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the (ROC) Curve
CAMUS	Cardiac Acquisitions for Multi-structure Ultrasound Segmentation (echocardiogram dataset)
CORAL	Correlation Alignment
CPEIC	Chaudhry Pervaiz Elahi Institute of Cardiology, Multan (dataset lineage to be confirmed)
CVD	Cardiovascular Disease
ECE	Expected Calibration Error
ECG	Electrocardiogram
Echo	Echocardiogram
ED / ES	End-Diastole / End-Systole
EDV / ESV	End-Diastolic Volume / End-Systolic Volume
EF	Ejection Fraction

Acronym	Definition
Grad-CAM	Gradient-weighted Class Activation Mapping
LV / LVEF	Left Ventricle / Left Ventricular Ejection Fraction
MAE	Mean Absolute Error
ONNX	Open Neural Network Exchange
PCP	Primary Care Provider
PR-AUC	Area Under the Precision–Recall Curve
SDS	Software Design Specification
SRS	Software Requirements Specification
SUS	System Usability Scale
TTA	Test-Time Adaptation
ARRDB	VitalDB Arrhythmia Database
DICOM	Digital Imaging and Communications in Medicine
DL	Deep Learning
FR / UC	Functional Requirement / Use Case
FYP	Final Year Project
GPU	Graphics Processing Unit
IoU	Intersection over Union
MI	Myocardial Infarction
JWT	JSON Web Token
RLS	Row-Level Security
REST	Representational State Transfer
ED	End-Diastole
ES	End-Systole
EDV	End-Diastolic Volume
ESV	End-Systolic Volume
EF	Ejection Fraction
Admin	Application administrator role
Clinician	Authenticated clinician role

Appendix H – Project Roles Summary

Table 28: Appendix H - Project Roles Summary

Team Member	Role	Responsibilities
Sahil Kumar	Data & ECG Modeling Lead	ECG dataset acquisition, preprocessing and patient-level splitting; ECG baseline training and generalization testing; ECG domain adaptation and active learning integration.
Jawad Ali	Echo Modeling & Compression Lead	Echo dataset acquisition and format conversion (CAMUS, EchoNet-Dynamic); echo baseline training and generalization testing; model compression and hardware benchmarking (both modalities).
Fatima	Integration & Product Lead	Backend development; frontend dashboards; on-demand Referral Summary Generation using FastAPI; react-driven summary-generation/review interface; integration with Supabase PostgreSQL; end-user testing; security/audit layer implementation.
All Members (shared)	Joint Responsibilities	Literature review and weekly decision-log updates; multimodal fusion network; proposal, mid-year report, final report and defense preparation. Referral summary data mapping, validation of generated summaries, and functional, security, and usability testing.
Dr. Sher Muhammad Daudpota	Supervisor	Project supervision, technical evaluation, and academic review.
Dr. Ismail Mangrio	Co-Supervisor	Technical guidance and review.

Appendix I – Open Issues (TBD)

Table 29: Appendix I - Open Issues (TDB)

ID	Open Issue	Related Section
OI-1	Accepted ECG/Echo file extensions, video codecs, MIME types, upload size limits, and validation policy are not specified.	3.2, UC-01, FR-23
OI-2	Final ECG-only triage mapping and fused model decision thresholds/category mapping are not fully specified.	UC-04, FR-24

OI-3	EF threshold boundaries conflict: methodology states EF <50% and EF ≤40%; prototype text describes >50% Normal, 40–50% Needs Review, and <40% Urgent Referral. Confirm exact treatment of EF=40% and EF=50%.	UC-04, FR-08, FR-24, 5.2
OI-4	Uncertainty threshold, calibration acceptance criteria, and handling of missing/incomplete modality inputs require final values.	FR-14 to FR-16, FR-24, 5.1–5.2
OI-5	User administration details, Admin capabilities, account lifecycle, password policy, session expiry, and account recovery behavior beyond Supabase defaults require confirmation.	FR-30 to FR-39, 5.3
OI-6	Case/file retention duration, deletion/export rights, storage bucket structure, signed URL strategy, and backup/restore expectations are not defined.	FR-27, FR-34 to FR-38, 5.3, Appendix A
OI-7	The proposal lists CPEIC as the local ECG dataset but also refers separately to a locally sourced ECG dataset. Confirm whether these are one dataset or two, and verify source, access, sample count, lead coverage, file format, labels, and clearance.	2.2, 2.7, Appendix B
OI-8	ARRDB input lead count, waveform format, sample rate, and compatibility with the primary 12-lead ECG model require dataset-level verification.	2.2, FR-01, Appendix B
OI-9	CPEIC ECG-image digitization feasibility, lead coverage, image quality, calibration, and label mapping are not confirmed.	FR-05, Appendix B
OI-10	MIMIC-IV-ECG and MIMIC-IV-Echo patient-level linkage, temporal matching window,	FR-18 to FR-20, Appendix B

	and the number of eligible paired records must be verified before fusion training.	
OI-11	The proposal's ED/ES frame identification and volume-estimation method, including geometric assumptions needed to derive EDV/ESV from segmentation, requires precise implementation documentation and validation.	FR-04, FR-07, Appendix D
OI-12	Deployment provider, frontend/backend resource tiers, environment configuration, domain, HTTPS certificate management, and production monitoring are not specified.	2.6, 3.2, Appendix A
OI-13	The final PostgreSQL schema, RLS policy definitions, storage policies, and backend service-role access pattern require design approval and security testing.	FR-27, FR-33 to FR-39, 5.3
OI-14	The proposal describes the usability evaluation as a simulation study; confirm recruitment, consent, task protocol, and whether the participants are volunteer primary-care providers.	2.2, FR-28, Appendix F
OI-15	Final referral summary template fields, required versus optional case information, and handling of unavailable analysis outputs must be confirmed during implementation.	3.2, UC-09, FR-41 to FR-48
OI-16	Summary versioning and regeneration behavior must be defined, including whether previous generated summaries are retained when a new summary is created after updated analysis results.	3.2, UC-09, FR-46, FR-49